

Bureau Veritas

Thruster Requirements

Comprehensive Surveyor Catalog

Based on NR467 Rules for Classification of Steel Ships

Sections 7, 14, and 15

Complete Plan Approval Reference Guide

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FOR BUREAU VERITAS PLAN APPROVAL SURVEYORS

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Document Purpose and Usage

This catalog provides a comprehensive overview of thruster requirements per Bureau Veritas Rules for Classification (NR467) and related guidance, designed as a practical reference for plan approval surveyors.

Key Features

- Complete drawing and document lists for all thruster types
- Explicit approval categories (**FA** or **FI**)
- Cross-references for overlapping requirements (steering gear, main shafting, hydraulics)
- Surveyor checklists by thruster type
- Testing and commissioning requirements
- Common submission errors and review tips

Document Structure

The catalog is organized into the following chapters:

1. **The Rule-Mandated Submissions (Authoritative)** — the verbatim rule tables
2. General Requirements (All Thrusters $\geq 110\text{kW}$)
3. Transverse Thrusters (Tunnel & Retractable)
4. Rotating and Azimuth Thrusters (incl. Podded Units)
5. Water-Jet Propulsion Systems
6. Voith-Schneider (Cycloidal) Propulsion Systems
7. Other Thruster Types (Rim-Driven, Pump-Jet, Gill/Jet, Steering Grid)
8. Prime Mover Considerations — Diesel-Engine vs. Electric-Motor Driven Thrusters
9. Thrusters as Part of Steering Gear
10. Thrusters as Part of Main Shafting
11. Hydraulic Systems & Pitch Control

12. Dynamic Positioning (DP) Thrusters
13. Ice Class & Cold-Climate (Winterisation) Requirements
14. Additional BV Class Notations Affecting Thruster Packages
15. Ship-Type & Service-Notation Specific Requirements
16. Testing & Commissioning
17. Surveyor Checklists by Type
18. Submission Cover Sheets by Ship Type (one-page tick-off)
19. Common Submission Errors
20. Quick Reference Tables
21. Regulatory References
22. Contact Information

How to Use This Catalog (Surveyor Workflow)

1. **Identify the thruster type** (Chapters 3–7) and confirm rated power. Thrusters ≥ 110 kW attract the full Sec. 15 submission set.
2. **Identify the prime mover** — electric motor or diesel engine (Chapter 8) — and add the corresponding drive-train drawings.
3. **Identify the functional role(s)**: propulsion, steering (Chapter 9), main shaft line (Chapter 10), or dynamic positioning (Chapter 12). Each role *adds* to the base package.
4. **Overlay additional notations & service constraints**: ice class / winterisation (Chapter 13), other BV additional class notations (Chapter 14), and the ship-type / service-notation requirements incl. SRtP, hazardous-area and offshore rules (Chapter 15).
5. **Verify hydraulics/pitch** (Chapter 11) and **testing & commissioning** (Chapter 16).
6. **Run the type-specific checklist** (Chapter 17) and screen against common errors (Chapter 19).

Navigating this PDF

This document is fully hyperlinked for quick reference:

- **Bookmarks panel** — open your PDF reader’s bookmarks/outline panel to jump to any chapter or section.
- **Table of contents** — every entry is clickable.
- **Cross-references** — “Chapter X”, “Section Y” and table references in the text are clickable links.
- **Search** — the text is fully selectable and searchable (e.g. search “MON-SHAFT”, “escort tug”, “ice”).

Legend

- **FA** = **For Approval** — corresponds to the rule’s category **A** (“to be submitted for approval”)
- **FI** = **For Information** — corresponds to the rule’s category **I** (“to be submitted for information”)
- **During Construction** = Certificates/reports generated during manufacturing

How to read this catalog — two tiers of content

Tier 1 — what the rule actually demands. Chapter 1 reproduces the *verbatim* submission tables of NR467 Sec. 15, Sec. 14, Part F and NR584, with the rule’s own **A/I** categorisation and an exact citation for every row. **That chapter is the authority. If anything elsewhere in this catalog conflicts with it, Chapter 1 governs.**

Tier 2 — what a surveyor examines to approve those items. Chapters 3–7 and the discipline chapters set out the engineering content behind each rule item: the calculations, drawings and analyses that a reviewer normally needs in order to be satisfied that (say) “Propeller” or “Rotating mechanism of the thruster” can be approved. **This is professional practice, not a rule-mandated document list.** A client cannot be required to submit a Tier 2 line as a separate deliverable purely because it appears here — cite the Tier 1 row it supports.

Note on rule references: Thruster requirements are found in **NR467, Part C, Chapter 1, Section 15** (“Thrusters”); steering gear in **Section 14**; shaft lines and propellers in **Sections 7–8**. Bracketed sub-references such as [2.3] are structural pointers within this catalog. Ice class, comfort, automation and DP requirements sit in **NR467 Part F** (Additional Class Notations, e.g. DYNAPOS for DP); service notations are in **Part D** (general) and **Part E** (offshore service vessels & tugs). Note **NR216** is the Materials & Welding rule note (not a DP rule). Always confirm the exact article against the rule edition applicable to the vessel’s contract/build date.

Chapter 1

The Rule-Mandated Submissions (Authoritative)

This chapter reproduces the *actual* submission tables of the rules, with the rule’s own **A/I** categorisation and an exact citation for every row. It was transcribed from Bureau Veritas Rules Explorer against **NR467 edition Jul 2026** and **NR584 edition Apr 2026**.

This chapter is the authority. Where any other chapter of this catalog lists a document that does not appear here, that document is *supporting technical content* (see the two-tier note in the Document Purpose chapter) — not a rule-mandated submission. Always confirm the exact article against the rule edition applicable to the vessel’s contract date at <https://rulesexplorer.bureauveritas.com/>.

1.1 Scope, definitions and thresholds

1.1.1 Power threshold

NR467 Pt C, Ch. 1, Sec. 15, [1.1.1] applies to thrusters developing power **equal to 110 kW or more**. A unit rated at exactly 110 kW *is* in scope. Below that, [1.1.2]: thrusters of less than 110 kW are to be built in accordance with sound marine practice and tested as required in [3.2] to the satisfaction of the Surveyor — the Sec. 15 plan tables are not applied.

1.1.2 What is (and is not) a thruster

Sec. 15 [1.2.1]: “A thruster is a propeller installed in a *revolving* nozzle or in a special transverse tunnel in the ship, or a water-jet. A thruster may be intended for propulsion, manoeuvring and steering or any combination thereof. **Propulsion propellers in fixed nozzles are not considered thrusters** (see Ch. 1, Sec. 8, [1.1.1]).”

Scope trap: a conventional propulsion propeller inside a *fixed* Kort nozzle is **not** a thruster. It is assessed under Sec. 8 (Propellers), not Sec. 15. Confirm whether the nozzle revolves before applying this catalog.

Other definitions: [1.2.2] a *transverse thruster* is an athwartship thruster developing thrust in a transverse direction for manoeuvring purposes; [1.2.3] an *azimuth thruster* can rotate through 360° to develop thrust in any direction; [1.2.4] a *water-jet* is a tubular casing (duct) enclosing an impeller, which may have means for deviating the jet to provide a steering function.

1.1.3 Number of units

Sec. 15 [1.3.1]: “In general, **at least two azimuth thrusters** are to be fitted in ships where these are the sole means of propulsion. Single azimuth thruster installations will be specially considered by the Society on a case by case basis.” This requirement *also applies to water-jets*.

1.1.4 Which tables apply

Per Sec. 15 [1.4]: athwartship and azimuth thrusters → **Tab. 1 + Tab. 2 + Tab. 3**; water-jets → **Tab. 4**; manufacturer data → **Tab. 5**, submitted together with the plans.

1.1.5 Duty changes the technical rule set

Table 1.1: Manoeuvring vs. propulsion — rule consequences

Aspect	Manoeuvring only	Propulsion / steering
Propeller [2.2.2]	Sec. 8 [2.5] without the 10% thickness increase	Sec. 8 [2.5] in full
Shaft [2.2.3]	Formula in Sec. 15 [2.2.3]; +10% in way of a keyed connection	Ch. 1, Sec. 7, [2.2.3]
Gears [2.2.4]	Sec. 6 with auxiliary gear safety factors	Sec. 6 with propulsion gear safety factors
Material tests	Need not be witnessed by a Surveyor, provided full test reports are made available — [3.1.2]	In the presence of a Surveyor , per Sec. 8 [4.1] — [3.1.1]
Ice class	Only Pt F, Ch. 8, Sec. 3, [3.5.1] (tunnel grids)	Full Pt F, Ch. 8
Steering gear	Not applicable	Sec. 14 applies, incl. Article 4

1.2 NR467 Sec.15 — Table 1: all thrusters

Table 1.2: NR467 Pt C, Ch. 1, Sec. 15, Tab.1 — Documentation to be submitted for all thrusters

No.	A/I	Documentation	Particulars
1	FI	General arrangement of the thruster	—
2	FA	Propeller	Including the applicable details mentioned in Ch. 1, Sec. 8
3	FA	Bearing details	—
4	FA	Propeller and intermediate shafts	—
5	FA	Gears	Including the applicable details mentioned in Ch. 1, Sec. 6

Note the categorisation: the *general arrangement* is FI (information), not FA. Earlier editions of this catalog listed it for approval; the rule does not.

1.3 NR467 Sec.15 — Table 2: transverse thrusters

Table 1.3: NR467 Pt C, Ch. 1, Sec. 15, Tab.2 — Documentation to be submitted for transverse thrusters

No.	A/I	Documentation	Particulars
1	FA	Structure of the tunnel	Showing the materials and their thickness
2	FA	Structural equipment or other connecting devices which transmit the thrust from the propeller to the tunnel	—
3	FA	Sealing devices	Propeller shaft gland and thruster-tunnel connection
4	FA	Pitch control device and corresponding monitoring system	For the adjustable pitch propellers

1.4 NR467 Sec.15 — Table 3: rotating and azimuth thrusters

Table 1.4: NR467 Pt C, Ch. 1, Sec. 15, Tab.3 — Documentation to be submitted for rotating and azimuth thrusters

No.	A/I	Documentation	Particulars
1	FA	Structural items	Nozzle, bracing, etc.
2	FA	Structural connection to hull	—
3	FA	Rotating mechanism of the thruster	—
4	FA	Thruster control system	—

No.	A/I	Documentation	Particulars
5	FA	Piping systems connected to thruster	—

1.5 NR467 Sec.15 — Table 4: water-jets

Table 1.5: NR467 Pt C, Ch. 1, Sec. 15, Tab. 4 — Document to be submitted for water-jets

No.	A/I	Documentation	Particulars
1	FI	General arrangement of the water-jet	—
2	FA	Casing (duct)	Showing: location and shape, materials, thicknesses, forces acting on the hull
3	FA	Details of the shafts, flanges, keys	—
4	FI	Sealing gland	—
5	FA	Bearings	—
6	FA	Impeller	—
7	FA	Steering and reversing buckets and their control devices as well as the corresponding hydraulic diagrams	—

1.6 NR467 Sec.15 — Table 5: manufacturer data

Table 1.6: NR467 Pt C, Ch. 1, Sec. 15, Tab. 5 — Document to be submitted for thrusters and water-jets

No.	A/I	Documentation	Particulars
1	FI	Data on ratings	Rated power, thrust and revolutions
2	FA	Material specifications of the major parts	Including their physical, chemical and mechanical properties
3	FA	Weldings	Where parts of thrusters are of welded construction. Including all particulars on the design of welded joints, welding procedures, heat treatments and non-destructive examinations after welding
4	FI	Background information on previous operating experience in similar applications	Where applicable

Note the categorisation: *data on ratings* — the “technical specification sheet” in commercial parlance — is **FI** (information), not **FA**.

1.7 Calculations demanded by the Sec.15 design articles

These are not in the tables, but the rule text of Article 2 expressly requires them to be submitted.

Table 1.7: Submissions required by the text of NR467 Sec. 15, Article 2

Reference	A/I	Requirement
Sec. 15 [2.2.5]	FA	Nozzle structure per Part B, Chapter 12 . Nozzle-to-hull scantlings and weld type/size specially considered; the Society <i>reserves the right to require a detailed stress analysis</i> for certain high power installations.
Sec. 15 [2.2.5]	FA	For steerable thrusters, the equivalent rudder stock diameter is to be calculated in accordance with Part B, Chapter 12.
Sec. 15 [2.2.6]	—	Transverse thruster tunnel thickness not less than the adjacent part of the hull . Tunnels connected to the hull by devices other than welding are specially considered.
Sec. 15 [2.3.2]	FA	Water-jet guide vanes and shaft supports: “Fatigue strength calculation is to be submitted.”
Sec. 15 [2.3.3]	FA	Water-jet stator/rotor: no natural frequency in the vicinity of the excitation frequencies due to stator–rotor hydrodynamic interaction. “Calculations are to be submitted for maximum speed and any currently used speed.”
Sec. 15 [2.3.5]	—	Water-jet steering performance and emergency steering availability at least equivalent to Sec. 14 [4.2] and [4.3] .
Sec. 15 [2.4.1]	—	Steering controls from the navigation bridge, the machinery control station and locally ; means to stop any running thruster at each control station; a thruster angle indicator at each steering control station, independent of the control system .

1.8 Minimum alarm and monitoring — Sec. 15 Table 6

Table 1.8: NR467 Pt C, Ch.1, Sec. 15, Tab.6 — Azimuth thrusters: minimum alarm and monitoring

Identification of system parameter	Alarm
Steering oil pressure	Low
Oil tank level	Low

These two lines are the *minimum* alarm and monitoring requirements the rule sets for propulsion and steering thrusters (see also Sec. 14 [4]). Anything beyond them is either an automation-notation requirement (Part F, Ch. 3) or good practice — not Sec. 15.

1.9 Testing and certification — Sec.15 Article 3

Table 1.9: NR467 Pt C, Ch. 1, Sec. 15, Article 3

Reference	Requirement
[3.1.1] Propulsion and steering thrusters	All materials intended for parts transmitting torque and for propeller/impeller blades are to be tested in accordance with Ch. 1, Sec. 8, [4.1] in the presence of a Surveyor.
[3.1.2] Transverse thrusters	Material testing for parts of athwartship thrusters does not need to be witnessed by a Surveyor, provided full test reports are made available to him.
[3.2.1] Thrusters	Thrusters are to be inspected as per the applicable requirements in Ch. 1, Sec. 8, [4.2].
[3.2.2] Prime movers	Prime movers are to be tested in accordance with the requirements applicable to the type of mover used.
[3.3.1] Certification	Thrusters are to be individually tested and certified by the Society.
[3.3.2] Mass produced	Mass produced thrusters of a type mentioned in [1.1] may be accepted within the framework of the type approval programme of the Society.

1.10 NR467 Sec.14 — thruster used as steering means

The steering-gear package for a thruster is *not* the whole of Sec.14 Table 1. The rule defines it as a specific subset:

NR467 Pt C, Ch. 1, Sec. 14, Tab. 1, item 13 (FI / **FA):** “For azimuth thrusters used as steering means: **Specification and drawings of the steering mechanism.** Where applicable, **documents 2 to 6 and 8 to 12** above.”

Table 1.10: Sec.14 Tab.1 rows pulled in by item 13

No.	A/I	Documentation	Particulars
13	FA	Specification and drawings of the steering mechanism	The azimuth-thruster entry itself
2	FI	General description of the installation and of its functioning principle	—
3	FI	Operating manuals of the steering gear and of its main components	—
4	FI	Description of the operational modes intended for steering in normal and emergency conditions	—
5	FA	For hydraulic steering gear, the schematic layout of the hydraulic piping of power actuating systems	Including the refilling system, design pressure, maximum working pressure expected in service, pipe diameter/thickness/material/connections, fluid tank capacity, flashpoint
6	FI	For hydraulic pumps of power units, the assembly longitudinal and transverse sectional drawings and the characteristic curves	—
8	FI	Constructional drawings of the relief valves for protection of the hydraulic actuators	Setting pressure; relieving capacity
9	FA	Diagrams of the electric power circuits	—
10	FA	Functional diagram of control, monitoring and safety systems	Including the remote control from the navigation bridge, with the location of control, monitoring and safety devices
11	FA	Constructional drawings of the strength parts providing a mechanical transmission of forces	—
12	FI	Calculation notes of the shrink-fit connections	—

Note: item 7 (rudder actuators) and item 1 (steering gear assembly drawing) are *not* in the range item 13 pulls in. Note 1 to Tab. 1 adds that constructional drawings are to be accompanied by the specification of materials and, where applicable, welding details and procedures.

1.10.1 Sec. 14 Table 2 — additional analyses

Table 1.11: NR467 Pt C, Ch. 1, Sec. 14, Tab. 2

No.	A/I	Documentation	When required
1	FA	Analysis in relation to the risk of single failure	Where required by Sec. 14 [2.4.2]
2	FA	Analysis in relation to the risk of hydraulic locking	Where required by Sec. 14 [2.4.5]
3	FA	Fatigue analysis and/or fracture mechanics analysis	Steering gear on oil tankers, chemical tankers or gas carriers — per Pt D, Ch. 7, Sec. 4, [7.2.2] and [7.3.1]

1.10.2 Sec. 14 Article 4 — performance criteria a surveyor must verify

Article 4 is titled “Design and construction — Requirements for ships equipped with thrusters as steering means”. Its numeric criteria are the heart of the steering review:

Table 1.12: NR467 Sec. 14, Article 4 — criteria

Reference	Criterion
[4.1.1]	The main and auxiliary steering gear may consist of azimuth thrusters, water-jets or cycloidal propellers complying with Sec. 15 as far as applicable.
[4.1.2]	Where the steering means consists of two or more thrusters , the control system is to include a device ensuring automatic synchronisation of the thruster rotation , unless each thruster is designed to withstand any additional forces resulting from the thrust exerted by the other thrusters.
[4.2.2]	Multiple steering-propulsion units: each unit to have a main and an auxiliary steering gear, or two or more identical steering actuating systems per [4.2.7]. Failure of one shall not render the other inoperative.
[4.2.3]	Single steering-propulsion unit: two or more steering actuating systems per [4.2.7], and a detailed risk assessment is to be submitted demonstrating that on any single failure in the steering gear, control system and power supply, ship steering is maintained.
[4.2.5]	Main steering: adequate strength and capable of steering at maximum ahead service speed (to be demonstrated); capable of changing direction from one side to the other at the declared steering angle limits at an average turning speed of not less than 2,3°/s with the ship running ahead at maximum ahead service speed; power operated for all ships ; so designed as not to be damaged at maximum astern speed (need not be proved by trials).

Reference	Criterion
[4.2.6]	Auxiliary steering: capable of steering at navigable speed and of being brought speedily into action; average turning speed not less than 0,5°/s with the ship running ahead at one half of maximum ahead service speed or 7 knots, whichever is the greater ; power operated where necessary to meet SOLAS II-1/29.4.2 and in any ship having more than 2 500 kW propulsion power per steering-propulsion unit .
[4.2.7]	Auxiliary steering gear may be omitted where the main steering gear has two or more identical power units and actuators and: on a <i>passenger ship</i> satisfies [4.2.5] with any one power unit out of operation; on a <i>cargo ship</i> satisfies [4.2.5] operating with all power units; and after a single failure in piping or one power unit, steering capability is maintained or speedily regained. Electric steering motors count as part of the power unit and actuator.
[4.2.8]	Where propulsion power exceeds 2 500 kW per thruster unit , an alternative power supply is to be provided automatically within 45 s , from the emergency source or an independent source located in the steering gear compartment (used only for this purpose), sufficient for the steering arrangements, their control system and the steering gear response indicator. Capacity at least 30 min for ships of 10 000 GT and upwards, at least 10 min for any other ship.
[4.3.1]	The use of water-jets as steering means will be given special consideration by the Society.

Declared steering angle limits are declared by the directional control system manufacturer for each ship-specific non-traditional steering means. Manoeuvrability tests (IMO Res. MSC.137(76)) are to be carried out at steering angles not exceeding those limits.

1.11 Electric drive — NR467 Pt C, Ch.2

Sec.15 [2.2.1] routes the drive as follows: **diesel engines** → Ch.1, Sec.2; **electric motors and their feeding systems** → **Ch.2, Sec.4**. Two express provisions apply to electric drives:

- provisions are to be made to **prevent starting of the motors whenever there are insufficient generators in operation**;
- **intermittent duty thrusters** will be the subject of special consideration by the Society.

1.11.1 Motors above 1 MW

NR467 Pt C, Ch.2, Sec.4, Article 5 — “Additional tests for rotating machines used as propulsion motor or thruster”. In addition to the standard Tab.1 tests, machines developing **more than 1 MW** are subject to the following during their assembly:

Table 1.13: NR467 Pt C, Ch.2, Sec.4, [5.1.1] — machines > 1 MW

Item	Requirement / test
Shaft line	Requirements of Ch.1, Sec.7 apply
Rotor winding assembly	Dynamic balancing
Stator winding assembly	Dielectric test (after impregnation); insulation resistance measurement (after impregnation)
Frame	Visual examination in compliance with design drawings; liquid penetrant test of 10% of the structure welds and 100% of the handling points
Watercooler	Visual examination in compliance with design drawings; performance test (temperature-rise measurement)
Hydrostatic jacking unit	Pressure test; working test under nominal conditions

1.12 Dynamic positioning — NR467 Pt F, Ch. 11, Sec. 5 (DYNAPOS)

Sec. 15 [1.1.1]: “For azimuth thrusters intended for dynamic positioning, the additional requirements in **Pt F, Ch. 11, Sec. 5** are to be complied with.” DYNAPOS [1.1.3] cross-refers back to Pt C, Ch. 1, Sec. 2 (engine-driven thrusters), **Pt C, Ch. 1, Sec. 15** (azimuthal and transverse thrusters) and Pt F, Chapter 3 (automation).

The FMEA is conditional. Table 1 item 17 (Failure Mode and Effect Analysis) is annotated “For symbols R and RS only”, and [2.4.1] confirms: “For installation intended to be assigned the notation DYNAPOS AM/AT R and DYNAPOS AM/AT RS an FMEA is to be carried out.” A plain DYNAPOS AM, AT or AM/AT installation does not attract an FMEA. Do not demand one as a matter of course.

Table 1.14: NR467 Pt F, Ch. 11, Sec. 5, Tab. 1 — DYNAPOS documentation

No.	A/I	Documentation	Particulars
1	FA	Documentation on the environment conditions long term distribution	See [9.4.1]
2	FI	Owner performance request	If any
3	FA	Capability plots	Diagram of the environment limit conditions for the conditions defined in the specification, and at least with all thrusters running and selected in DP, and the worst case failure
4	FA	Functional block diagram for the sensor and reference systems	Reference systems for position / environmental conditions
5	FA	Functional block diagram for the control unit	—
6	FA	Single line diagram and specification of the cables between the different equipment units	Power, control, display
7	FA	Electrical load balance	—
8	FA	List of the equipment units	Manufacturer’s identification, type and model for each
9	FA	Type test reports	For the sensors of the measurement systems, or equivalent
10	FA	Type test report or case-by-case approval certificate	For the DP control system; tests to be in accordance with Pt C, Ch. 3, Sec. 6
11	FA	For approval of propulsion based on rotary azimuth thrusters: layout drawings of thrust units, thrust shafts and blocks; arrangement of hull passages; thrust curves of each propulsion unit	—
Continued on next page			

No.	A/I	Documentation	Particulars
12	FA	Electrical power management layout drawings and specification	If provided on board
13	FA	Internal communication system description	—
14	FA	Description of the operator stations	Layout on board; descriptive diagrams of the display consoles
15	FA	Alarm list and parameter values displayed on the consoles	—
16	FA	Program of FMEA proving tests alongside quay and at sea	—
17	FA	Failure mode and effect analysis (FMEA)	For symbols R and RS only — see [2.4]
18	FA	Study of possible interaction between thrusters	—
19	FI	Technical specification of the positioning system	—
20	FI	Operating manual of the positioning system	Including description of the equipment, maintenance guide, emergency procedures
21	FI	Vessel-specific DP-operation manuals	See [7.1]
22	FA	Program of FMEA annual tests alongside quay and at sea	—

1.13 Ice class — NR467 Pt F, Ch. 8

1.13.1 The notations

Pt F, Ch. 8, Sec. 1, [1.1.1] assigns: **ICE CLASS IA SUPER, ICE CLASS IA, ICE CLASS IB, ICE CLASS IC, ICE CLASS ID, YOUNG ICE 1, YOUNG ICE 2**. Per [1.1.2], the strengthening requirements — *excepting* ICE CLASS ID, YOUNG ICE 1 and YOUNG ICE 2 — are equivalent to the **Finnish-Swedish Ice Class Rules 2019, as amended**, for ships trading in the Baltic in winter or equivalent ice conditions.

1.13.2 The scope split — this is where packages are most often over-demanded

NR467 Sec. 15 [1.1.1], verbatim: “Thrusters intended for propulsion and steering of ships with an ICE CLASS notation are to comply with the additional requirements of Part F, Chapter 8. **Transverse thrusters intended for manoeuvring of ships with an ICE CLASS notation are required to comply with additional requirement Pt F, Ch. 8, Sec. 3, [3.5.1] only.**”

And Pt F, Ch. 8, Sec. 3, [3.5.1] in its entirety reads:

“Tunnels of transverse thrusters are to be fitted with grids for protection against ice impacts.”

That is the whole ice requirement for a manoeuvring bow or stern tunnel thruster on an ice-class ship: a protective grid. Do not apply the propulsion-machinery ice package to it.

1.13.3 Propulsion and steering thrusters — the full package

Pt F, Ch. 8, Sec. 3, Article 1 applies to propulsion machinery for ICE CLASS IA SUPER / IA / IB / IC. [1.1.1] states that the regulations “*also apply to azimuthing and fixed thrusters for main propulsion, considering loads resulting from propeller-ice interaction and loads on the thruster body/ice interaction*”.

Table 1.15: Pt F, Ch. 8, Sec. 3, Article 1 — propulsion machinery in ice

Sub-article	Content
[1.4]	Design ice loads on propeller blades; axial design loads for open and ducted propellers; torsional design loads; blade failure load
[1.5]	Calculation of blade stresses; acceptability criterion; fatigue design of the propeller blade; acceptability criterion for fatigue
[1.6]	Design of blade flange and bolts, propeller hub and controllable pitch mechanism; blade bolts and flanges; components of the pitch control system; servo oil pressure
[1.7]	Shaft line: design principle, materials, scantling of propeller, intermediate and thrust shafts; scantling of gear shaft; calculation of propeller blade failure; alternative design procedure
[1.8]	Couplings: flange, shrunk, keyed, flexible; clutches
[1.9]	Gears: application factor, inertia ratio, safety factors
[1.11]	Azimuthing thruster body: design principles; extreme ice impact loads; extreme ice loads on the thruster hull when penetrating an ice ridge; acceptability criterion for static loads; thruster body global vibration; steering gear design torque in ice; protection of the steering gear against excessive torque

Sub-article	Content
[3.6.1]	Test and certification of propellers: Pt C, Ch. 1, Sec. 8, [4] applies; additionally material tests per [1.3] and [2.2.1]

1.14 NR584 — Azimuth thrusters in ice for polar class ships and icebreakers

A separate rule note — easily missed. NR584 (edition Apr 2026) applies to ships fitted with an **azimuth propulsion system** and assigned a service notation **icebreaker** or an additional class notation **POLAR CLASS**. It is a complete rule note in its own right, with 8 sections and its own submission tables, and it is substantially more demanding than NR467 Sec. 15. If your vessel is polar-class with azimuth propulsion, *this* is the controlling document.

1.14.1 Application and structure

Per NR584 Sec. 1, [1.1.2] the azimuth propulsion system may be:

- a **podded electrical thruster** — steering unit, slewing bearing, rudder part of the system, and the pod containing the electric motor; or
- a **Z/L geared thruster** — steering unit, upper gearing (Z-drive), lower gearing, nozzle, and the prime mover (diesel engine or electric motor).

The eight sections are: Sec. 1 General; Sec. 2 Structure of the Thruster Body; Sec. 3 Machinery Assessment; Sec. 4 Electrical Installations; Sec. 5 Steering Units; **Sec. 6 Structural Assessment of the Thruster Body in Ice**; Sec. 7 Tests; Sec. 8 Certification.

1.14.2 Submission tables

NR584 Sec. 1, [1.3] requires **Table 2** (55 items, below), plus **Table 3** for solid propellers *or* **Table 4** for built-up and controllable pitch propellers, plus **Table 5** manufacturer data.

Table 1.16: NR584 Sec. 1, Tab. 2 — documentation to be submitted

No.	A/I	Documentation	Particulars
1	FI	General arrangement of the thruster	Including cross sections and sealing arrangements exposed to sea
2	FI	General layout of the thruster	—
3	FA	Azimuth thruster general specification	Rating, operation modes, limitations and maximum design loads (torque, thrust, hydrodynamic loads, loads transferred to hull, steering loads and cycles); material specifications for principal parts/assemblies; welding procedures
4	FI	Environmental conditions	—
5	FA	Failure Mode and Effects Analysis of the propulsion	For the cooling, lubrication, ventilation systems, sealing system, steering functions of the thruster, thruster control system and fire safety
6	FA	Process and instrumentation of azimuth thruster	Including all the auxiliary services
Continued on next page			

No.	A/I	Documentation	Particulars
7	FA	Specification and data sheet of the auxiliary services	—
8	FA	Arrangements and details of the auxiliary services	—
9	FA	Diagrams of cooling, ventilation, draining/bilge, lubricating and hydraulic systems	—
10	FI	Description of the operational modes intended for steering in normal and emergency conditions	—
11	FI	Operation manual of the thruster in ice conditions	—
12	FA	Structural drawings of the thruster	—
13	FA	Structural connection to hull	—
14	FA	Arrangement of the azimuth propulsion system supports	—
15	FA	Ice load calculation of the thruster body	—
16	FA	Stress calculation for the thruster body	—
17	FA	Details of structural items (nozzle, bracing, etc.)	Including information on clearance between propeller and nozzle
18	FI	Maximum propulsive power in ice conditions	As a function of thruster steering angle and shaft rotational speed
19	FA	Propeller details	Including the applicable details mentioned in [1.3.2]
20	FA	Detailed drawing of the propeller shaft	Including the brake
21	FA	Propeller and intermediate shafts	—
22	FI	Installation drawing of the bearings	—
23	FA	Bearing details	—
24	FI	Bearing data	Maximum allowable static and dynamic load capacity, permissible rotational speed range, axial displacement, operating conditions
25	FI	Dynamic load calculation of the bearings	—
26	FI	Lifetime calculation of the bearings	—
27	FA	Seals drawings	—
28	FA	Gears details	Including the applicable details required by the Society
29	FA	Piping systems connected to thruster	—

Continued on next page

No.	A/I	Documentation	Particulars
30	FI	Assembly longitudinal and transverse sectional drawings of hydraulic pumps of power units, and their characteristic curves	—
31	FA	General arrangement of electrical installations	—
32	FA	Specification and data sheet of main electrical components	—
33	FA	Schematic electrical diagrams of main electrical components	—
34	FA	General arrangement diagram of main electrical components	—
35	FA	Thruster control system	—
36	FA	Diagram of the supply for monitoring and control systems of propulsion motors	—
37	FA	Diagram of the supply, monitoring and control systems of controllable pitch propellers	—
38	FA	Principles of control system and its power supply	—
39	FA	Alarm and monitoring system details	List of alarms and monitoring points; power supply diagram
40	FA	Safety system details	List of monitored parameters for safety system; power supply diagram
41	FA	Test program for electrical propulsion motors and electrical slip rings	—
42	FA	Diagrams of the electric power circuits	—
43	FA	Functional diagram of control, monitoring and safety systems	Including the remote control from the navigating bridge, with the location of control, monitoring and safety devices
44	FI	Operating manuals of the steering unit and of its main components	—
45	FI	General arrangement of the steering system	Main components (steering bearing and steering drive), protection and locking devices
46	FI	Assembly drawing of the steering unit	Including sliding blocks, guides, stops and other similar components
47	FI	General description of the steering unit installation and of its functioning principle	—
48	FA	Detailed drawings of the main components of the steering system	Including locking device and all load carrying components
Continued on next page			

No.	A/I	Documentation	Particulars
49	FA	Rotating mechanism of the thruster	—
50	FA	For hydraulic steering unit, the schematic layout of the hydraulic piping of power actuating systems	Including refilling system; design pressure; maximum working pressure expected in service; pipe diameter, thickness, material and connections; fluid tank capacity; flashpoint
51	FA	Assembly drawings of the actuators and constructional drawings of their components	For hydraulic actuators: design torque, maximum working pressure, relief valve setting pressure
52	FI	Constructional drawings of the relief valves for protection of the hydraulic actuators	Setting pressure; relieving capacity
53	FA	Diagram of the supply, monitoring and control systems of the steering unit	—
54	FA	Risk analysis of single failure	Where the main steering unit comprises two or more identical power units
55	FA	Risk analysis of hydraulic locking	Where more than one steering unit can be simultaneously operated

1.14.3 NR584 propeller and manufacturer tables

Table 1.17: NR584 Sec.1, Tab.3 (solid propellers) and Tab.4 (built-up and CP propellers)

No.	A/I	Documentation
Tab. 3 — solid propellers		
1	FA	Sectional assembly
2	FA	Blade and hub details
3	FI	Rating (power, rpm, etc.)
4	FA	Data and procedures for fitting propeller to the shaft
Tab. 4 — built-up and controllable pitch propellers (items 1–4 as above, plus)		
5	FA	Blade bolts and pre-tensioning procedures
6	FI	Pitch corresponding to maximum propeller thrust and to normal service condition
7	FA	Pitch control mechanism
8	FA	Pitch control hydraulic system

Table 1.18: NR584 Sec.1, Tab.5 — additional data and documentation to be submitted by manufacturer

No.	A/I	Documentation	Particulars
1	FI	Onboard installation procedure	—
2	FI	Rated power and revolutions	—

No.	A/I	Documentation	Particulars
3	FI	Rated thrust	—
4	FA	Material specifications of the major parts	Including their physical, chemical and mechanical properties
5	FA	Particulars on the design of welded joints, welding procedures, heat treatments and non-destructive examinations after welding	Where parts of thrusters are of welded construction
6	FI	Background information on previous operating experience in similar applications	Where applicable

Chapter 2

General Requirements - All Thrusters $\geq 110\text{kW}$

2.1 Applicability

All thrusters developing power equal to 110 kW or more ($\geq 110\text{ kW}$) must comply with NR467 Section 15 requirements (NR467 Pt C, Ch.1, Sec.15, [1.1.1]). Thrusters of less than 110 kW are to be built in accordance with sound marine practice and tested per Sec.15 [3.2] to the satisfaction of the Surveyor (NR467 Pt C, Ch.1, Sec.15, [1.1.2]) — verify with surveyor for the exact reduced scope.

2.2 Two Axes to Classify Before Listing Drawings

Before applying any type checklist, fix two attributes of the unit — they change *which* rules and drawings apply as much as the thruster type does:

2.2.1 Duty: manoeuvring vs. propulsion

- **Manoeuvring (auxiliary)** — e.g. a bow/stern tunnel thruster used only for low-speed manoeuvring. Assessed under Sec. 15 as auxiliary machinery; it is *not* part of the main shaft line.
- **Propulsion (main)** — e.g. an azimuth on a tug, a podded unit, a water-jet or a Voith-Schneider that propels the ship. The propeller, shaft and gear are classed as **main propulsion**: the main-shaft-line requirements of Chapter 10 apply, a **torsional vibration calculation is mandatory**, and barred/continuous-speed ranges must be confirmed.
- **Take-me-home / emergency propulsion** — a retractable or auxiliary unit providing get-home capability; assess independence from the main train (cross-reference redundant-propulsion notation, Chapter 14).

The same physical thruster therefore attracts a heavier package when it is the ship's propulsion than when it is purely for manoeuvring.

2.2.2 Propeller configuration: open vs. ducted (nozzle)

- **Open (non-ducted) propeller** — blade and hub strength, cavitation and the usual shaft/bearing package.
- **Ducted (nozzle) propeller** — *adds* the nozzle/duct structure and strength drawing, nozzle profile/geometry with propeller-tip clearance, and the nozzle-to-unit/hull connection. Distinguish:
 - **Fixed nozzle** (Kort-type) — structural item only; improves bollard pull.
 - **Steering nozzle** — the nozzle also steers, so the **steering-gear package** of Chapter 9 applies (actuator, control, redundancy, SOLAS II-1/29 where it is the ship's means of steering).

Surveyor note: record duty and propeller configuration on the submission cover sheet (Chapter 18). The interactive Quick-Reference tool builds the exact drawing list once type, duty, propeller configuration, prime mover, role, ship type and notations are selected.

2.3 Mandatory Submissions for All Thruster Types

Read with Chapter 1. The rule-mandated list for every thruster is **Sec. 15 Tab. 1** (Table 1.2) plus **Tab. 5** (Table 1.6) — nine rows in total. The table below expands those rows into the engineering content a reviewer normally needs; it is *supporting practice*, and the A/I categories shown against the two rule rows have been aligned to the rule.

Table 2.1: General Requirements - All Thrusters

#	Drawing/Document	Category	Content Requirements	Reference
1	General Arrangement Drawing	FI	• Thruster location on vessel • Installation orientation • Access provisions • Clearances to other equipment • Coordinate system reference	Sec.15, Tab.1 #1
2	Data on ratings (technical specification sheet)	FI	• Rated power, thrust and revolutions (the rule's own wording) • Operating voltage/pressure • Weight and dimensions • Operating depth/pressure rating	Sec.15, Tab.5 #1
Continued on next page				

Table 2.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
3	Material specifications of the major parts	FA	• Physical, chemical and mechanical properties (the rule's own wording) • Material grades for all major components • Corrosion protection details • Material certificates (during construction) — NR216 / NR266	Sec.15, Tab.5 #2
4	Weldings	FA	• Design of welded joints, welding procedures, heat treatments and NDE after welding (rule wording, where parts are of welded construction) • WPS / PQR and welder qualification records • Weld inspection plan	Sec.15, Tab.5 #3; NR216
5	Manufacturer's Quality Plan	FI	• Manufacturing sequence • Inspection and test points • Quality control procedures • Traceability system	<i>Practice</i> — supports Sec.15 [3.3.1] certification
6	Operation & Maintenance Manual	FI	• Operating instructions • Maintenance schedule • Troubleshooting guide • Safety precautions	<i>Practice</i> (rule-required only for ice: NR584 Tab.2 #11)
7	Spare Parts List	FI	• Recommended spare parts • Critical component identification • Part numbers and suppliers	<i>Practice</i> — not a Sec.15 table row
8	Installation Manual	FI	• Installation procedure • Alignment requirements • Foundation specifications • Connection details	<i>Practice</i> (rule-required only for ice: NR584 Tab.5 #1)

Continued on next page

Table 2.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
9	Background information on previous operating experience	FI	• Service record of the same or similar unit • Where applicable	Sec.15, Tab.5 #4

Chapter 3

Transverse Thrusters (Tunnel & Retractable)

Rule basis for this chapter. The rule-mandated submissions for a transverse thruster are **Sec. 15 Tab. 1** (Table 1.2, 5 rows) + **Tab. 2** (Table 1.3, 4 rows) + **Tab. 5** (Table 1.6, 4 rows) — **13 rows in total**. Everything below expands those rows into the engineering content a reviewer examines. It is *supporting practice*, not an independent list of deliverables a client can be required to produce.

Note also **Sec. 15 [2.2.6]**: the tunnel thickness is not to be less than the adjacent part of the hull; and for a manoeuvring-only unit the propeller need not carry the 10% thickness increase, the shaft follows the Sec. 15 [2.2.3] formula, and gears use *auxiliary* gear safety factors.

3.1 Tunnel Thrusters (Fixed Installation)

3.1.1 Structural Drawings & Calculations

Table 3.1: Tunnel Thrusters - Structural Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Tunnel Structural Arrangement	FA	• Tunnel diameter and length • Tunnel position relative to baseline/centerline • Plate thickness and material • Reinforcement arrangement • Connection to hull structure	NR467 Sec.15, [2.1]
2	Tunnel Structural Calculations	FA	• Pressure loads (internal/external) • Structural stress analysis • Buckling analysis • Fatigue assessment • FEA results (if applicable)	NR467 Sec.15, [2.1]
Continued on next page				

Table 3.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
3	Tunnel End Grating/Grid Design	FA	• Grating dimensions and material • Attachment method • Structural strength verification • Debris protection	NR467 Sec.15, [2.2]
4	Hull Penetration Details	FA	• Opening dimensions • Reinforcement around opening • Welding details • Watertight integrity provisions	NR467 Part B, Ch.5

3.1.2 Thruster Unit - Mechanical Components

Table 3.2: Tunnel Thrusters - Mechanical Components

#	Drawing/Document	Category	Content Requirements	Reference
5	Thruster Unit General Assembly	FA	• Complete unit cross-section • Component identification • Material specifications • Assembly sequence	NR467 Sec.15, [2.3]
6	Propeller Design Drawing	FA	• Propeller diameter and pitch • Number of blades • Blade profile and geometry • Hub design • Material specification	NR467 Sec.15, [2.4]
7	Propeller Strength Calculations	FA	• Blade root stress analysis • Hub stress analysis • Centrifugal force calculations • Fatigue life assessment • Cavitation analysis	NR467 Sec.15, [2.4]
8	Propeller Shaft Drawing	FA	• Shaft diameter and length • Material specification • Surface finish requirements • Keyway details • Seal locations	NR467 Sec.15, [2.5]

Continued on next page

Table 3.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
9	Shaft Strength Calculations	FA	• Torsional strength analysis • Bending stress calculations • Combined stress analysis • Deflection calculations • Critical speed analysis • Fatigue assessment	NR467 Sec.15, [2.5]; Sec.7
10	Bearing Arrangement Drawing	FA	• Bearing type and manufacturer • Bearing housing design • Lubrication arrangement • Seal design • Cooling provisions (if applicable)	NR467 Sec.15, [2.6]
11	Bearing Load Calculations	FA	• Radial and axial loads • Bearing life calculation (L10) • Lubrication requirements • Temperature analysis	NR467 Sec.15, [2.6]
12	Gearbox Assembly Drawing	FA	• Gear arrangement (bevel, helical, etc.) • Gear ratios • Casing design • Lubrication system • Mounting arrangement	NR467 Sec.15, [2.7]
13	Gear Tooth Calculations	FA	• Gear tooth strength (ISO 6336) • Contact stress analysis • Bending stress analysis • Lubrication film thickness • Efficiency calculations	NR467 Sec.15, [2.7]
14	Gearbox Lubrication System	FA	• Oil type and capacity • Circulation system (if forced) • Cooling arrangement • Filtration system • Oil level and temperature monitoring	NR467 Sec.15, [2.7]

3.1.3 Drive System (Electric or Hydraulic)

For Electric Motor Drive

Table 3.3: Tunnel Thrusters - Electric Drive System

#	Drawing/Document	Category	Content Requirements	Reference
15	Electric Motor Specification	FA	• Motor type (induction, synchronous) • Rated power and speed • Voltage and frequency • Insulation class • Protection class (IP rating) • Cooling method • Mounting arrangement	NR467 Sec.15, [2.8]; IEC 60092-301
16	Motor Foundation Drawing	FA	• Foundation structure design • Vibration isolation details • Bolt arrangement • Alignment provisions • Structural attachment to hull	NR467 Sec.15, [2.8]
17	Motor-Gearbox Coupling	FA	• Coupling type and manufacturer • Alignment tolerances • Torque capacity • Installation procedure	NR467 Sec.15, [2.8]

For Hydraulic Motor Drive

Table 3.4: Tunnel Thrusters - Hydraulic Drive System

#	Drawing/Document	Category	Content Requirements	Reference
18	Hydraulic Motor Specification	FA	• Motor type (axial piston, radial piston, gear) • Displacement and speed range • Rated pressure • Efficiency curves • Mounting details	NR467 Sec.15, [2.9]
19	Hydraulic System Schematic	FA	• Complete hydraulic circuit • Component symbols and identification • Pressure and flow ratings • See Chapter 11 for detailed requirements	NR467 Sec.15, [2.9]

3.1.4 Control System

Table 3.5: Tunnel Thrusters - Control System

#	Drawing/Document	Category	Content Requirements	Reference
20	Control System Block Diagram	FA	- Control architecture - Input/output signals - Control logic overview - Interface with ship systems	NR467 Sec.15, [2.10]; Part C, Ch.4
21	Control Panel Layout	FA	- Panel arrangement - Component locations - Wiring diagram - Protection devices	NR467 Sec.15, [2.10]
22	Electrical Single-Line Diagram	FA	- Power supply arrangement - Circuit breakers and protection - Emergency stop circuit - Monitoring and alarm systems	NR467 Sec.15, [2.10]; IEC 60092-504
23	Control Software Description	FA	- Software functionality - Control algorithms - Safety functions - Software validation documentation	NR467 Part C, Ch.4

3.1.5 Additional Calculations & Documentation

Table 3.6: Tunnel Thrusters - Additional Calculations

#	Drawing/Document	Category	Content Requirements	Reference
24	Thrust Calculation	FA	- Theoretical thrust calculation - Efficiency factors - Operating curves (thrust vs. speed) - Bollard pull prediction	NR467 Sec.15, [2.11]
25	Vibration Analysis	FA	- Natural frequency calculations - Forced vibration analysis - Resonance avoidance verification - Vibration isolation design	NR467 Sec.15, [2.12]

Continued on next page

Table 3.6 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
26	Noise Level Prediction	FA	• Airborne noise prediction • Structure-borne noise analysis • Compliance with IMO noise code • Noise reduction measures	NR467 Sec.15, [2.12]
27	Thermal Analysis	FA	• Heat generation calculations • Cooling requirements • Temperature rise predictions • Ventilation requirements	NR467 Sec.15, [2.13]

3.2 Retractable Thrusters

3.2.1 Additional Requirements Beyond Fixed Tunnel Thrusters

All submissions from Section 2.1 apply, PLUS the following:

Table 3.7: Retractable Thrusters - Additional Requirements

#	Drawing/Document	Category	Content Requirements	Reference
28	Retraction Mechanism Assembly	FA	• Complete mechanism design • Lifting/lowering system • Guide rail arrangement • Travel limits and stops • Position indicators	NR467 Sec.15, [2.14]
29	Hydraulic Cylinder Specifications	FA	• Cylinder bore and stroke • Operating pressure • Rod diameter and material • Seal specifications • Mounting arrangement	NR467 Sec.15, [2.14]
30	Retraction System Calculations	FA	• Lifting force requirements • Hydraulic pressure calculations • Cylinder sizing verification • Retraction/extension time calculations • Dynamic load analysis	NR467 Sec.15, [2.14]
31	Hull Opening & Closure Design	FA	• Opening dimensions • Closure plate/door design • Sealing arrangement (extended position) • Sealing arrangement (retracted position) • Locking mechanism	NR467 Sec.15, [2.15]; Part B, Ch.5
32	Hull Opening Structural Analysis	FA	• Local hull strength calculations • Reinforcement design around opening • Stress concentration analysis • Fatigue assessment • FEA results (typically required)	NR467 Sec.15, [2.15]; Part B, Ch.5
Continued on next page				

Table 3.7 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
33	Watertight Integrity Documentation	FA	• Seal design and material • Pressure testing requirements • Leakage criteria • Compliance with subdivision requirements	NR467 Part B, Ch.5
34	Locking System Design	FA	• Mechanical locks (extended position) • Retracted position locks • Locking force calculations • Fail-safe provisions • Interlock system	NR467 Sec.15, [2.16]
35	Position Indication System	FA	• Position sensors (type and location) • Indication on bridge/control panel • Alarm system (intermediate positions) • Redundancy provisions	NR467 Sec.15, [2.16]
36	Emergency Lowering System	FA	• Emergency power source • Manual lowering provisions • Emergency procedure • Time to deploy	NR467 Sec.15, [2.17]
37	Hydraulic System for Retraction	FA	• HPU specifications • Control valve arrangement • Piping layout • Accumulator sizing (if applicable) • Emergency lowering circuit • See Chapter 11 for detailed requirements	NR467 Sec.15, [2.17]
38	Retraction/Extension Procedures	FI	• Step-by-step operating procedure • Pre-operation checks • Safety precautions • Troubleshooting	NR467 Sec.15, [2.18]
39	Emergency Recovery Procedures	FI	• Emergency lowering procedure • Manual operation instructions • Recovery from failure modes • Contact information for support	NR467 Sec.15, [2.18]

Chapter 4

Rotating and Azimuth Thrusters

Rule basis for this chapter. The rule-mandated submissions for a rotating or azimuth thruster are **Sec. 15 Tab. 1** (Table 1.2) + **Tab. 3** (Table 1.4) + **Tab. 5** (Table 1.6) — **14 rows in total**. If the unit is also the ship's steering means, add the Sec. 14 Tab. 1 item 13 subset (Section 1.10) and verify the Article 4 criteria (Section 1.10.2). Everything below is the supporting engineering content behind those rows.

Nozzle and hull connection — Sec. 15 [2.2.5]: the nozzle structure is assessed under **Part B, Chapter 12**; nozzle-to-hull scantlings and weld type/size are specially considered and the Society may require a detailed stress analysis for certain high power installations. For *steerable* thrusters the **equivalent rudder stock diameter** is calculated per Part B, Chapter 12.

If POLAR CLASS or icebreaker: NR584 governs — see Section 1.14.

4.1 Azimuth Thrusters (Fixed Pitch & Controllable Pitch)

4.1.1 General Arrangement & Structure

Table 4.1: Azimuth Thrusters - General Arrangement

#	Drawing/Document	Category	Content Requirements	Reference
1	General Arrangement Drawing	FA	• Thruster location on vessel • Installation orientation • Azimuth rotation range (typically 360°) • Clearance envelope (rotating) • Access for maintenance	NR467 Sec.15, [3.1]
2	Azimuth Unit Complete Assembly	FA	• Longitudinal and transverse sections • All major components identified • Material specifications • Assembly sequence • Lifting points	NR467 Sec.15, [3.1]
Continued on next page				

Table 4.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
3	Hull Penetration & Foundation	FA	• Hull opening dimensions • Structural reinforcement • Foundation design • Watertight integrity details • Load transfer to hull structure	NR467 Sec.15, [3.2]; Part B, Ch.5
4	Hull Structure Calculations	FA	• Local hull strength analysis • Foundation load distribution • Stress analysis • Fatigue assessment • FEA results (typically required)	NR467 Sec.15, [3.2]; Part B, Ch.5

4.1.2 Propeller & Shaft System

For Fixed Pitch Propellers

Table 4.2: Azimuth Thrusters - Fixed Pitch Propeller

#	Drawing/Document	Category	Content Requirements	Reference
5	Propeller Design Drawing	FA	• Propeller diameter and pitch • Number of blades • Blade geometry (sections) • Hub design • Material specification • Blade attachment method	NR467 Sec.15, [3.3]
6	Propeller Strength Calculations	FA	• Blade stress analysis • Hub stress analysis • Centrifugal forces • Hydrodynamic forces • Fatigue life assessment • Cavitation analysis	NR467 Sec.15, [3.3]

For Controllable Pitch Propellers (ADDITIONAL to Fixed Pitch)

Table 4.3: Azimuth Thrusters - Controllable Pitch Mechanism

#	Drawing/Document	Category	Content Requirements	Reference
7	CP Hub Mechanism Assembly	FA	• Hub internal mechanism • Pitch control rod/linkage • Blade retention system • Bearing arrangement • Seal design	NR467 Sec.15, [3.4]
8	Blade Retention Calculations	FA	• Blade root connection strength • Centrifugal force analysis • Hydrodynamic moment analysis • Bolt/pin strength verification • Fatigue assessment	NR467 Sec.15, [3.4]
9	Pitch Control Mechanism	FA	• Hydraulic cylinder (in hub or external) • Control rod design • Linkage system • Pitch range and indicators • Fail-safe position	NR467 Sec.15, [3.4]; Ch.11
10	Pitch Control Hydraulic System	FA	• Hydraulic supply arrangement • Rotary joint/slip ring design • Control valve system • Pitch feedback system • See Chapter 11 for detailed requirements	NR467 Sec.15, [3.4]; Ch.11

4.1.3 Vertical Shaft & Column

Table 4.4: Azimuth Thrusters - Vertical Shaft & Column

#	Drawing/Document	Category	Content Requirements	Reference
11	Vertical Shaft Assembly	FA	• Shaft dimensions and material • Upper and lower bearing locations • Thrust bearing arrangement • Seal system • Coupling details	NR467 Sec.15, [3.5]; Sec.7
Continued on next page				

Table 4.4 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
12	Vertical Shaft Calculations	FA	• Torsional strength • Bending stress (from propeller forces) • Combined stress analysis • Deflection calculations • Critical speed analysis • Fatigue assessment	NR467 Sec.15, [3.5]; Sec.7
13	Thrust Bearing Design	FA	• Bearing type and manufacturer • Load capacity • Lubrication system • Cooling arrangement • Monitoring provisions	NR467 Sec.15, [3.6]
14	Thrust Bearing Calculations	FA	• Axial load analysis • Bearing life calculation • Temperature rise analysis • Lubrication film thickness	NR467 Sec.15, [3.6]
15	Radial Bearing Arrangement	FA	• Upper and lower bearing design • Bearing housing • Lubrication system • Seal arrangement	NR467 Sec.15, [3.7]
16	Column Structure Design	FA	• Column/strut dimensions • Material specification • Hydrodynamic fairing • Structural strength • Corrosion protection	NR467 Sec.15, [3.8]
17	Seal System Design	FA	• Seal type and material • Seal arrangement (upper/lower) • Pressure rating • Leakage monitoring • Replacement procedure	NR467 Sec.15, [3.9]

4.1.4 Gear System

Table 4.5: Azimuth Thrusters - Gear System

#	Drawing/Document	Category	Content Requirements	Reference
18	Gear Train Assembly	FA	• Gear arrangement (bevel, helical, planetary) • Gear ratios • Gear casing design • Lubrication system • Access for inspection	NR467 Sec.15, [3.10]
19	Gear Tooth Strength Calculations	FA	• Tooth bending stress (ISO 6336) • Contact stress • Pitting resistance • Scuffing analysis • Efficiency calculations	NR467 Sec.15, [3.10]
20	Gear Lubrication System	FA	• Oil type and capacity • Circulation system (forced/splash) • Cooling arrangement • Filtration system • Oil level and temperature monitoring	NR467 Sec.15, [3.10]

4.1.5 Slewing (Azimuth) System

Table 4.6: Azimuth Thrusters - Slewing System

#	Drawing/Document	Category	Content Requirements	Reference
21	Slewing Bearing Design	FA	• Bearing type and manufacturer • Internal/external gear (if applicable) • Load capacity (axial, radial, moment) • Mounting arrangement • Lubrication provisions	NR467 Sec.15, [3.11]
22	Slewing Bearing Calculations	FA	• Load analysis (static and dynamic) • Bearing life calculation • Bolt preload calculations • Overturning moment analysis	NR467 Sec.15, [3.11]
Continued on next page				

Table 4.6 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
23	Azimuth Drive Mechanism	FA	• Drive motor (electric or hydraulic) • Gear/pinion arrangement • Drive ratio • Brake system • Locking mechanism (if required)	NR467 Sec.15, [3.12]
24	Azimuth Drive Calculations	FA	• Torque requirements • Acceleration/deceleration analysis • Motor sizing verification • Brake sizing • Response time calculations	NR467 Sec.15, [3.12]
25	Position Feedback System	FA	• Position sensor type and location • Accuracy and resolution • Signal processing • Redundancy (if required for DP/steering)	NR467 Sec.15, [3.13]

4.1.6 Power Transmission

For Electric Drive

Table 4.7: Azimuth Thrusters - Electric Power Transmission

#	Drawing/Document	Category	Content Requirements	Reference
26	Electric Motor Specification	FA	• Motor type and rating • Voltage and frequency • Cooling method • Insulation and protection class • Mounting arrangement	NR467 Sec.15, [3.14]; IEC 60092-301
27	Power Cable Routing	FA	• Cable route from switchboard • Cable support and protection • Penetrations through bulkheads • Cable sizing calculations	NR467 Sec.15, [3.14]; IEC 60092-352
Continued on next page				

Table 4.7 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
28	Slip Ring or Rotary Joint	FA	• Type and manufacturer • Current rating • Maintenance requirements • Cooling provisions	NR467 Sec.15, [3.14]

For Hydraulic Drive

Table 4.8: Azimuth Thrusters - Hydraulic Power Transmission

#	Drawing/Document	Category	Content Requirements	Reference
29	Hydraulic Motor Specification	FA	• Motor type and displacement • Rated pressure and speed • Efficiency curves • Mounting details	NR467 Sec.15, [3.15]
30	Hydraulic System Schematic	FA	• Complete hydraulic circuit • HPU specifications • Piping arrangement • Control valve system • See Chapter 11 for detailed requirements	NR467 Sec.15, [3.15]; Ch.11

4.1.7 Control System

Table 4.9: Azimuth Thrusters - Control System

#	Drawing/Document	Category	Content Requirements	Reference
31	Control System Architecture	FA	• Overall control system topology • Propulsion control • Azimuth control • Pitch control (CP thrusters) • Interface with ship systems	NR467 Sec.15, [3.16]; Part C, Ch.4
32	Bridge Control Console	FA	• Control panel layout • Joystick/lever arrangement • Display and indication • Alarm system • Emergency stop provisions	NR467 Sec.15, [3.16]

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Table 4.9 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
33	Local Control Station	FA	- Local control panel design - Control mode selection - Indication and monitoring - Interlock with bridge control	NR467 Sec.15, [3.16]
34	Control Software Description	FA	- Software architecture - Control algorithms - Safety functions - Software validation documentation - Version control	NR467 Part C, Ch.4
35	Electrical Single-Line Diagram	FA	- Power supply arrangement - Circuit breakers and protection - Emergency power provisions - Monitoring and alarm circuits	NR467 Sec.15, [3.16]; IEC 60092-504

4.1.8 Additional Calculations & Documentation

Table 4.10: Azimuth Thrusters - Additional Calculations

#	Drawing/Document	Category	Content Requirements	Reference
36	Thrust and Torque Calculations	FA	- Propeller thrust curves - Torque requirements - Power absorption - Efficiency analysis - Bollard pull prediction	NR467 Sec.15, [3.17]
37	Vibration Analysis	FA	- Natural frequency calculations - Forced vibration analysis - Resonance avoidance - Foundation vibration isolation - Acceptable vibration levels	NR467 Sec.15, [3.18]
38	Noise Analysis	FA	- Airborne noise prediction - Structure-borne noise analysis - Compliance with noise regulations - Noise reduction measures	NR467 Sec.15, [3.18]
Continued on next page				

Table 4.10 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
39	Hydrodynamic Analysis	FA	• Propeller hydrodynamic design • Cavitation analysis • Efficiency optimization • Interaction effects with hull	NR467 Sec.15, [3.19]
40	Thermal Analysis	FA	• Heat generation in gears, bearings, seals • Cooling system capacity • Temperature rise predictions • Ventilation requirements	NR467 Sec.15, [3.20]

4.2 Podded Propulsion Units

Podded propulsion units include all requirements from Section 3.1 (Azimuth Thrusters), with the following **ADDITIONAL** or **MODIFIED** requirements:

4.2.1 Pod-Specific Structural Components

Table 4.11: Podded Propulsion - Pod Structure

#	Drawing/Document	Category	Content Requirements	Reference
41	Pod Housing Design	FA	- Pod shell structure - Material specification - Hydrodynamic fairing - Structural strength analysis - Corrosion protection system	NR467 Sec.15, [3.21]
42	Pod Housing Calculations	FA	- Pressure loading analysis - Structural stress analysis - Fatigue assessment - Impact resistance - FEA results	NR467 Sec.15, [3.21]
43	Strut Structure Design	FA	- Strut dimensions and material - Hydrodynamic profile - Connection to pod and hull - Internal cable/pipe routing - Access provisions	NR467 Sec.15, [3.22]
44	Strut Structural Calculations	FA	- Bending stress analysis - Torsional stress - Combined loading - Fatigue analysis - Vibration analysis	NR467 Sec.15, [3.22]

4.2.2 Electric Motor in Pod

Table 4.12: Podded Propulsion - In-Pod Motor

#	Drawing/Document	Category	Content Requirements	Reference
45	In-Pod Motor Design	FA	• Motor type (permanent magnet, induction) • Rated power and speed • Voltage and frequency • Insulation system (marine environment) • Rotor and stator assembly • Bearing arrangement	NR467 Sec.15, [3.23]; IEC 60092-301
46	Motor Thermal Design	FA	• Heat generation calculations • Cooling system design • Temperature limits • Thermal monitoring • Insulation life assessment	NR467 Sec.15, [3.23]
47	Motor Cooling System	FA	• Cooling method (water-cooled, oil-cooled) • Heat exchanger design • Coolant circulation system • Pump specifications • Temperature monitoring	NR467 Sec.15, [3.24]
48	Motor Mounting & Support	FA	• Motor mounting arrangement in pod • Alignment provisions • Vibration isolation • Thermal expansion accommodation	NR467 Sec.15, [3.23]

4.2.3 Power Supply Through Strut

Table 4.13: Podded Propulsion - Power Supply

#	Drawing/Document	Category	Content Requirements	Reference
49	Power Cable Routing	FA	• Cable route through strut • Cable support and protection • Cable entry into pod • Sealing arrangements • Cable sizing calculations	NR467 Sec.15, [3.25]; IEC 60092-352
Continued on next page				

Table 4.13 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
50	Slip Ring or Rotary Transformer	FA	• Type and manufacturer • Current and voltage rating • Cooling provisions • Maintenance access • Reliability data	NR467 Sec.15, [3.25]
51	Power Distribution in Pod	FA	• Internal switchgear/junction boxes • Protection devices • Monitoring systems • Watertight/waterproof rating	NR467 Sec.15, [3.25]

4.2.4 Propeller Direct Drive

Table 4.14: Podded Propulsion - Direct Drive

#	Drawing/Document	Category	Content Requirements	Reference
52	Direct Drive Arrangement	FA	• Motor shaft to propeller connection • Coupling or direct mounting • Bearing arrangement • Seal system • Alignment procedure	NR467 Sec.15, [3.26]
53	Shaft Seal System	FA	• Seal type and material • Seal arrangement • Pressure rating • Leakage monitoring • Emergency sealing provisions	NR467 Sec.15, [3.26]

4.2.5 Pod-Specific Testing

Table 4.15: Podded Propulsion - Testing Requirements

#	Drawing/Document	Category	Content Requirements	Reference
54	Pressure Testing Procedure	FA	• Test pressure and duration • Test setup • Acceptance criteria • Leakage limits	NR467 Sec.15, [3.27]
Continued on next page				

Table 4.15 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
55	Motor Insulation Testing	FA	• Insulation resistance tests • High voltage tests • Partial discharge tests • Test conditions and acceptance criteria	IEC 60092-301

Chapter 5

Water-Jet Propulsion Systems

Rule basis for this chapter. Water-jets have their *own* table: **Sec. 15 Tab. 4** (Table 1.5, 7 rows), submitted per [1.4.2], plus **Tab. 5** manufacturer data (Table 1.6). Note that Tab. 4 places the *general arrangement* and the *sealing gland* in category **FI**.

Two calculations are demanded by the rule text itself: **[2.3.2]** a *fatigue strength calculation* for guide vanes and shaft supports; and **[2.3.3]** a *stator/rotor blade natural-frequency calculation* for maximum speed and any currently used speed, demonstrating no natural frequency near the stator-rotor hydrodynamic interaction excitation.

[2.3.5]: steering performance and emergency steering availability at least equivalent to Sec. 14 [4.2] and [4.3]. Note Sec. 14 [4.3.1]: *the use of water-jets as steering means will be given special consideration by the Society*. **[1.3.1]** (at least two units where they are the sole means of propulsion) applies to water-jets as well.

5.1 General Arrangement

Table 5.1: Water-Jets - General Arrangement

#	Drawing/Document	Category	Content Requirements	Reference
1	Water-Jet General Arrangement	FA	• Installation location on vessel • Inlet and outlet positions • Nozzle orientation • Access for maintenance • Clearances	NR467 Sec.15, [4.1]
2	Hull Integration Drawing	FA	• Inlet duct design • Hull penetration details • Structural integration • Hydrodynamic considerations • Reinforcement arrangements	NR467 Sec.15, [4.2]; Part B, Ch.5

5.2 Water-Jet Unit Components

Table 5.2: Water-Jets - Unit Components

#	Drawing/Document	Category	Content Requirements	Reference
3	Water-Jet Unit Assembly	FA	• Complete unit cross-section • Pump assembly • Inlet duct • Nozzle and steering/reversing mechanism • Material specifications	NR467 Sec.15, [4.3]
4	Impeller Design	FA	• Impeller geometry • Number of blades • Material specification • Blade profile • Hub design	NR467 Sec.15, [4.4]
5	Impeller Strength Calculations	FA	• Blade stress analysis • Hub stress analysis • Centrifugal forces • Hydrodynamic forces • Fatigue assessment • Cavitation analysis	NR467 Sec.15, [4.4]
6	Pump Shaft Design	FA	• Shaft dimensions and material • Bearing locations • Seal arrangements • Coupling details	NR467 Sec.15, [4.5]
7	Pump Shaft Calculations	FA	• Torsional strength • Bending stress • Combined stress analysis • Deflection • Critical speed • Fatigue assessment	NR467 Sec.15, [4.5]; Sec.7
8	Bearing Arrangement	FA	• Bearing types and locations • Bearing housing • Lubrication system • Seal design • Cooling provisions	NR467 Sec.15, [4.6]
9	Bearing Load Calculations	FA	• Radial and axial loads • Bearing life calculation • Lubrication requirements • Temperature analysis	NR467 Sec.15, [4.6]

5.3 Inlet Duct System

Table 5.3: Water-Jets - Inlet Duct

#	Drawing/Document	Category	Content Requirements	Reference
10	Inlet Duct Design	FA	• Duct geometry and dimensions • Material specification • Structural design • Hydrodynamic profile • Debris protection (grating/screen)	NR467 Sec.15, [4.7]
11	Inlet Duct Calculations	FA	• Structural strength analysis • Pressure loading • Flow analysis • Cavitation assessment	NR467 Sec.15, [4.7]
12	Hull Penetration & Reinforcement	FA	• Opening dimensions • Reinforcement design • Watertight integrity • Structural attachment	NR467 Sec.15, [4.7]; Part B, Ch.5

5.4 Steering and Reversing Mechanism

Table 5.4: Water-Jets - Steering & Reversing

#	Drawing/Document	Category	Content Requirements	Reference
13	Steering Nozzle Design	FA	• Nozzle geometry • Deflection range • Actuator arrangement • Structural design • Material specification	NR467 Sec.15, [4.8]
14	Reversing Bucket Design	FA	• Bucket geometry • Deployment mechanism • Actuator system • Structural strength • Material specification	NR467 Sec.15, [4.9]
15	Steering/Reversing Actuators	FA	• Hydraulic cylinder specifications • Operating pressure • Force requirements • Position feedback • Emergency provisions	NR467 Sec.15, [4.10]
Continued on next page				

Table 5.4 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
16	Steering/Reversing Calculations	FA	• Actuator force requirements • Structural strength of nozzle/bucket • Response time analysis • Hydraulic system sizing	NR467 Sec.15, [4.10]

5.5 Drive System

5.5.1 For Direct Drive from Main Engine

Table 5.5: Water-Jets - Direct Drive

#	Drawing/Document	Category	Content Requirements	Reference
17	Drive Shaft Arrangement	FA	• Shaft line from engine to water-jet • Bearing locations • Coupling arrangements • Alignment provisions	NR467 Sec.15, [4.11]; Sec.7
18	Drive Shaft Calculations	FA	• Torsional strength • Bending stress • Critical speed analysis • Alignment calculations • See Chapter 10 for full requirements	NR467 Sec.15, [4.11]; Sec.7

5.5.2 For Gearbox Drive

Table 5.6: Water-Jets - Gearbox Drive

#	Drawing/Document	Category	Content Requirements	Reference
19	Gearbox Assembly	FA	• Gear arrangement • Gear ratios • Casing design • Lubrication system • Mounting arrangement	NR467 Sec.15, [4.12]

Continued on next page

Table 5.6 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
20	Gear Calculations	FA	• Gear tooth strength • Contact stress • Bending stress • Lubrication analysis • Efficiency	NR467 Sec.15, [4.12]

5.6 Hydraulic System (for steering/reversing)

Table 5.7: Water-Jets - Hydraulic System

#	Drawing/Document	Category	Content Requirements	Reference
21	Hydraulic System Schematic	FA	• Complete hydraulic circuit • HPU specifications • Control valves • Piping arrangement • See Chapter 11 for detailed requirements	NR467 Sec.15, [4.13]; Ch.11

5.7 Control System

Table 5.8: Water-Jets - Control System

#	Drawing/Document	Category	Content Requirements	Reference
22	Control System Architecture	FA	• Control system topology • Steering control • Reversing control • Engine/pump speed control • Interface with ship systems	NR467 Sec.15, [4.14]; Part C, Ch.4
23	Bridge Control Console	FA	• Control panel layout • Joystick/lever arrangement • Display and indication • Alarm system • Emergency stop	NR467 Sec.15, [4.14]
24	Electrical Single-Line Diagram	FA	• Power supply • Protection devices • Monitoring circuits • Emergency power provisions	NR467 Sec.15, [4.14]; IEC 60092-504

5.8 Performance Calculations

Table 5.9: Water-Jets - Performance Calculations

#	Drawing/Document	Category	Content Requirements	Reference
25	Thrust and Performance Calculations	FA	• Thrust curves • Power absorption • Efficiency analysis • Speed predictions • Bollard pull	NR467 Sec.15, [4.15]
26	Cavitation Analysis	FA	• Cavitation inception speed • Cavitation patterns • Erosion risk assessment • Mitigation measures	NR467 Sec.15, [4.15]
27	Vibration Analysis	FA	• Natural frequency calculations • Forced vibration analysis • Resonance avoidance • Foundation design	NR467 Sec.15, [4.16]
28	Noise Analysis	FA	• Airborne noise prediction • Structure-borne noise • Compliance with regulations • Noise reduction measures	NR467 Sec.15, [4.16]

Chapter 6

Voith-Schneider (Cycloidal) Propulsion Systems

Rule basis. NR467 Sec. 15 does not define a cycloidal propeller, so there is no dedicated table. The rule does, however, expressly recognise cycloidal propellers as an acceptable steering means: **Sec. 14 [4.1.1]** — “The main and auxiliary steering gear . . . may consist of thrusters of the following types: azimuth thrusters, water-jets, **cycloidal propellers**, complying with the provisions of Ch. 1, Sec. 15, *as far as applicable*.” Apply the rotating/azimuth tables (Tab. 1 + Tab. 3 + Tab. 5) as the closest fit and **agree the scope with the Society in writing before review**.

6.1 Description & Classification Basis

The Voith-Schneider Propeller (VSP), or cycloidal propeller, is a vertical-axis propulsor in which a set of vertical blades projects below a rotating casing (rotor). Each blade oscillates about its own vertical axis through a kinematic control gear linked to a movable control (steering) centre. By displacing the control centre, the unit produces thrust of *any magnitude in any horizontal direction* at constant rotor speed. A VSP therefore combines **propulsion and steering in a single unit**; the steering function makes **NR467 Part C, Ch 1, Sec 14 (Steering Gear) and SOLAS II-1 Reg. 29 requirements applicable** (see Chapter 9). Typical applications: Voith Water Tractors, double-ended ferries, buoy tenders, mine-countermeasure vessels.

VSP units are assessed under **NR467 Pt C, Ch 1, Sec 15** for the mechanical thruster elements, **Sec 14** for the steering/control function, and **Sec 7** where the drive forms part of the propulsion shaft line.

6.2 VSP-Specific Structural & Mechanical Components

Table 6.1: Voith-Schneider Propeller — Mechanical & Structural Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	VSP Unit General Assembly	FA	• Complete unit cross-section (rotor casing, blades, gear) • Number of blades and blade circle diameter • Rotor speed and orbit geometry • Material list and lifting arrangement	NR467 Sec.15, [5.1]
2	Blade (Wing) Design Drawing	FA	• Blade profile, chord, span • Blade shaft dimensions and material • Blade root / shaft connection • Surface finish and corrosion protection	NR467 Sec.15, [5.2]
3	Blade Shaft & Blade Strength Calculations	FA	• Hydrodynamic blade loads (thrust + side force) • Bending + torsion of blade shaft • Centrifugal loads at max rotor speed • Fatigue assessment (reversing loads)	NR467 Sec.15, [5.2]; Sec.7
4	Blade Bearing Arrangement	FA	• Upper/lower blade bearings, seals • Bearing load and L10 life • Lubrication path from oil system	NR467 Sec.15, [5.3]
5	Kinematic Control Gear (Steering Mechanism)	FA	• Control-centre / eccentric arrangement • Control rods, links, crank geometry • Servo actuator interface • Strength of linkage members and pins • Fail-safe / neutral-thrust behaviour	NR467 Sec.15, [5.4]; Sec.14

Continued on next page

Table 6.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
6	Rotor Casing & Main Bevel Gear	FA	• Rotor casing structure and thrust path • Input bevel gear (horizontal drive → vertical rotor) • Gear tooth calculations (ISO 6336) • Main thrust bearing design and life • Lubrication and oil monitoring	NR467 Sec.15, [5.5]
7	Bottom Cover / Guard Plate Structure	FA	• Guard plate below blade circle • Structural strength (grounding/impact) • Attachment to unit and hull • Detachment / fail-safe considerations	NR467 Sec.15, [5.6]; Part B, Ch.5
8	Hull Seat, Foundation & Well Structure	FA	• Foundation ring and bolting • Local hull reinforcement / FEA • Watertight integrity of installation well • Load transfer (thrust, steering moment)	NR467 Sec.15, [5.6]; Part B, Ch.5
9	Seal System (Rotor & Blade Shafts)	FA	• Seal type, arrangement and material • Leakage monitoring / oil-in-water prevention • EAL fluid if CLEANSHIP (see Ch.14)	NR467 Sec.15, [5.7]

6.3 VSP Drive, Servo Control & Steering Function

Table 6.2: Voith-Schneider Propeller — Drive & Control

#	Drawing/Document	Category	Content Requirements	Reference
10	Prime Mover Interface	FA	• Diesel engine or electric motor input (see Ch.8) • Input coupling / clutch / cardan shaft • Torsional vibration calculation of the complete drive train	NR467 Sec.15; Sec.7
11	Servo / Hydraulic Control Unit	FA	• Control (steering) hydraulic schematic • Two independent power/servo systems where used for steering (redundancy) • Pressure ratings, relief, accumulators • See Chapter 11	NR467 Sec.14; Ch.11
12	Steering Control System & Bridge Console	FA	• Control transmitter (tiller/joystick) to control centre • Thrust magnitude & direction indication • Follow-up / non-follow-up control • Alarms, emergency stop, mode changeover	NR467 Sec.14; Part C, Ch.4
13	Steering Function Compliance Package	FA	• FMEA of steering function • Single-failure / duplication analysis • Response time, hard-over to hard-over • SOLAS II-1 Reg.29 demonstration (see Ch.9)	NR467 Sec.14; SOLAS II-1/29

Surveyor note (VSP): A VSP is simultaneously a propeller, a gear unit *and* a steering gear. Do not approve the mechanical package in isolation — confirm that the steering-function documentation of Chapter 9 (redundancy, control stations, alarms, response time, SOLAS compliance) has also been submitted. The blade guard/bottom plate and its hull attachment are frequently under-documented; require structural substantiation against grounding loads.

Chapter 7

Other Thruster Types (Rim-Driven, Pump-Jet, Gill/Jet, Steering Grid)

No dedicated rule tables exist for these types. NR467 Sec. 15 [1.2] defines only three things: the *transverse thruster*, the *azimuth thruster* and the *water-jet*. The unit types in this chapter must be mapped onto the closest applicable table — rim-driven and steering-grid units onto Tab. 3 (rotating/azimuth), pump-jet and gill/jet units onto Tab. 4 (water-jet) — and **the scope agreed with the Society in writing before review**. Do not present the lists below as rule requirements. Remember the scope exclusion of [1.2.1]: a propulsion propeller in a *fixed* nozzle is **not** a thruster and is assessed under Sec. 8.

This chapter addresses less-common propulsor arrangements recognised by BV. Each follows the **general submissions of Chapter 2** plus the requirements of the *nearest analogous conventional type*, with the additional type-specific items below. Confirm the intended notation and service (propulsion, take-me-home, DP, steering) before fixing the package.

7.1 Rim-Driven Thrusters (Integrated Motor Propulsors)

Rim-driven / hubless thrusters integrate a permanent-magnet (or induction) motor whose rotor forms the propeller-blade tip ring inside the duct. There is no mechanical shaft line or reduction gear; excitation and speed are set by a frequency converter. Assessment shifts from gearing toward **electrical machine, converter and duct-bearing** substantiation.

Table 7.1: Rim-Driven Thruster — Type-Specific Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Integrated Motor & Duct Assembly	FA	- Motor type (PM / induction), pole/winding data - Rotor rim / blade-tip ring construction - Duct structure and stator support - Cooling (seawater / freshwater path)	NR467 Sec.15; Pt C, Ch.2; IEC 60092-301
Continued on next page				

Table 7.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
2	Frequency Converter / Drive Package	FA	• Converter rating and topology • Harmonic / power-quality analysis • Protection, cooling, EMC • See Chapter 8 (electric drive)	NR467 Pt C, Ch.2; IEC 60092-101
3	Bearing & Seal Arrangement	FA	• Water-lubricated / magnetic bearing design • Bearing load and wear-life basis • Gap / air-gap control, seal design	NR467 Sec.15, [6.1]
4	Blade & Rim Strength Calculations	FA	• Blade + rim stress (hydro + magnetic + centrifugal) • Fatigue and corrosion allowance • Cavitation assessment	NR467 Sec.15, [6.1]; Sec.7

7.2 Pump-Jet & Ducted Rotor-Stator Propulsors

Pump-jet units (ducted rotor with stator vanes, often for low-signature or shallow-draught craft) are treated as a ducted propeller + shaft line. Submit the **duct/stator structure and strength, rotor blade design and strength, shaft line (Chapter 10), bearings, seals** and the relevant drive package. Where the unit also steers (deflector/vectoring), add the steering function package of Chapter 9.

7.3 Gill / Hydro-Jet Transverse Thrusters (Pump-Type)

Pump-type transverse thrusters draw water through a hull inlet and discharge through athwartship outlets, providing side thrust without an open tunnel. Submit as for a tunnel thruster (Chapter 3) but replace the tunnel structure items with: **inlet grating and duct, internal pump/impeller design and strength, discharge outlet and directional-valve arrangement, hull penetration and watertight integrity**, plus the pump drive package.

7.4 Steering Grid / Steering-Nozzle & Rudder-Propeller Variants

Fixed-nozzle (Kort-type) units with a steering grid or a steerable nozzle behave as combined propulsion + steering devices. Submit the **nozzle/grid structure and strength, propeller in nozzle, shaft line, steering actuator and control**, and the full steering-gear compliance package of Chapter 9 (SOLAS II-1 Reg.29 where the device is the ship's means of steering).

Surveyor note (other types): For any novel or hybrid propulsor, apply the *function-based* approach — decompose the unit into propeller/impeller, structure, shaft line, gear (if any), drive

(electric/diesel/hydraulic), bearings/seals and control — and require the corresponding chapter's submissions for each function. Novel technology may additionally require an Approval-in-Principle (AiP) or risk-based/alternative-design assessment.

Chapter 8

Prime Mover Considerations — Diesel-Engine vs. Electric-Motor Driven Thrusters

8.1 Purpose

The preceding type chapters split the drive into *electric* or *hydraulic*. This chapter consolidates the **additional package elements driven by the choice of prime mover** so the surveyor can confirm the correct drive-train documentation is present. A thruster may be driven by:

- an **electric motor** (constant- or variable-speed, converter-fed), typical of diesel-electric ships, DP vessels, cruise ships and pods; or
- a **diesel engine** (direct/geared mechanical drive), typical of tugs, OSVs and workboats with inboard engines driving Z-drive azimuth or tunnel thrusters via shafting, clutches and gearboxes; or
- a **hydraulic motor** (covered per type chapter and Chapter 11), common on retractable and smaller units.

8.2 Electric-Motor-Driven Thrusters — Additional Submissions

Table 8.1: Electric-Motor Prime Mover — Additional Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Propulsion/Drive Motor Data Sheet	FA	• Type (induction / PM / synchronous) • Rated & overload power, torque-speed curve • Voltage, frequency, insulation & IP class • Cooling method, temperature rise (IEC 60034)	NR467 Pt C, Ch.2; IEC 60092-301
Continued on next page				

Table 8.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
2	Frequency Converter (VFD) Specification & Arrangement	FA	• Converter topology and rating • Output torque ripple at thruster • Protection, redundancy, bypass • Cooling and heat load to space	NR467 Pt C, Ch.2
3	Harmonic & Power-Quality Analysis	FA	• THD at main switchboard (limits per rules) • Effect on generators/other consumers • Mitigation (multi-pulse, active front end, filters)	NR467 Pt C, Ch.2
4	Supply Transformer & Switchboard Interface	FA	• Transformer rating, vector group, cooling • Feeder protection and discrimination • Short-circuit & earth-fault co-ordination	NR467 Pt C, Ch.2; IEC 60092-201
5	Power Cable Sizing & Routing	FA	• Current rating, voltage drop, derating • Routing, support, bulkhead penetrations • Segregation (DP redundancy where relevant)	IEC 60092-352 / -352
6	Motor Protection, Monitoring & Cooling	FA	• Winding/bearing temperature monitoring • Overload, earth-fault, differential protection • Space heaters, forced cooling, condensation control	NR467 Pt C, Ch.2 / Ch.3
7	Torque-Ripple / Torsional Interaction Note	FA	• Converter harmonics vs. shaft-line torsionals • Barred-speed / avoidance ranges if any • Cross-reference shaft TVC (Chapter 10)	NR467 Sec.7
8	Integration with PMS / Blackout Recovery	FI	• Load-dependent start/stop, load shedding • Blackout and thruster restart sequence	NR467 Pt C, Ch.3

8.3 Diesel-Engine-Driven Thrusters — Additional Submissions

Where a thruster (typically a Z-drive azimuth on a tug/OSV, or a directly driven tunnel/CP unit) is driven by a diesel engine, the **engine and its auxiliary systems become part of the approvable package** under **NR467 Pt C, Ch 1, Sec 2 (Diesel Engines)** in addition to the thruster under Sec 15. A **torsional vibration calculation (TVC) of the complete engine-clutch-shafting-thruster system is mandatory.**

Table 8.2: Diesel-Engine Prime Mover — Additional Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Diesel Engine Approval / Type Data	FA	• Engine general arrangement & rating (MCR) • Type Approval / unit certification status • Crankshaft & main running-gear data • Safety devices (overspeed, LO low-pressure shutdown)	NR467 Pt C, Ch.1, Sec.2
2	Torsional Vibration Calculation (TVC)	FA	• Complete engine-coupling-gear-thruster model • Barred/critical speed ranges • Stresses in shafts, coupling, gear at criticals • Clutch-in/out and misfiring conditions	NR467 Sec.2; Sec.7
3	Clutch & Flexible Coupling	FA	• Clutch type, torque & engagement control • Flexible coupling stiffness/damping data • Alignment and vibration decoupling	NR467 Sec.2; Sec.7
4	Engine Foundation & Chocking	FA	• Seating structure and holding-down bolts • Resilient mounts / chocking arrangement • Alignment provisions and thrust transmission	NR467 Sec.2; Part B, Ch.5
Continued on next page				

Table 8.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
5	Fuel Oil System	FA	• FO service/settling tanks, transfer, filtration • Piping, valves, quick-closing valves • Leakage containment, high-pressure line shielding	NR467 Pt C, Ch.1 (Piping)
6	Lubricating Oil System	FA	• LO pumps, cooler, filters, sump • Pressure/temperature monitoring & alarms	NR467 Pt C, Ch.1
7	Cooling Water System	FA	• Jacket & charge-air cooling circuits • Sea-water / central cooling, pumps, expansion	NR467 Pt C, Ch.1
8	Starting & Control Air / Starting System	FA	• Air receivers, compressors, capacity for starts • Or electric-start battery capacity	NR467 Pt C, Ch.1
9	Exhaust Gas System	FA	• Piping, expansion, lagging, silencer • Back-pressure, boundary insulation (fire) • SCR/EGR arrangement if fitted	NR467 Pt C, Ch.1; SOLAS II-2
10	Engine Control, Safety & Monitoring	FA	• Governor / speed control, over-speed trip • Alarms & safeguards, local/remote control • Integration with thruster & bridge controls	NR467 Pt C, Ch.1 / Ch.3
11	Emissions / NOx Documentation	FI	• EIAPP certificate & NOx Technical File • MARPOL Annex VI Tier compliance	MARPOL Annex VI
12	Engine-Room Ventilation & Air Supply	FI	• Combustion air and space ventilation • Heat balance / temperature rise	NR467 Pt C, Ch.1

Surveyor note (prime mover): The single most common gap for diesel-driven Z-drive tugs is a **missing or incomplete TVC for the whole train** (engine + clutch + cardan/shaft + azimuth

gears + propeller). For electric drives, the recurring gaps are the **harmonic/power-quality study** and **cable-sizing/segregation**. Hybrid PTI/PTO arrangements must document *every* driving path and the mode-changeover logic. Cross-reference the shaft-line torsional/alignment package in Chapter 10.

Chapter 9

Thrusters as Part of Steering Gear

9.1 Applicability

NR467 Pt C, Ch. 1, Sec. 14, [1.1.1]: the requirements of Sec. 14 apply to the steering gear systems of all mechanically propelled ships, “**and to the steering mechanism of thrusters used as means of propulsion**”.

Sec. 14 [4.1.1] states which thruster types may constitute the main and auxiliary steering gear: **azimuth thrusters, water-jets and cycloidal propellers**, complying with Sec. 15 as far as applicable.

The thruster steering package is a defined subset — not the whole of Sec. 14 Table 1. The rule entry is **Sec. 14, Tab. 1, item 13**: “For azimuth thrusters used as steering means: *Specification and drawings of the steering mechanism*. Where applicable, *documents 2 to 6 and 8 to 12* above.” Items 1 and 7 are **not** pulled in. The exact subset is reproduced in Table 1.10; the additional analyses are Table 1.11; and the numeric performance criteria a surveyor must verify are Table 1.12.

Do not simply demand the full rudder steering-gear package.

The key criteria from **Sec. 14, Article 4** (full text in Section 1.10.2), which surveyors most often need at the desk:

- **Main steering:** average turning speed $\geq 2,3^\circ/\text{s}$ at maximum ahead service speed, within the declared steering angle limits; power operated for all ships.
- **Auxiliary steering:** $\geq 0,5^\circ/\text{s}$ at half maximum ahead service speed *or 7 knots, whichever is greater*; power operated where needed for SOLAS II-1/29.4.2 and in any ship with $> 2\,500\text{ kW}$ per steering-propulsion unit.
- **Single unit:** two or more steering actuating systems *plus a detailed risk assessment* ([4.2.3]).
- **Multiple units:** main + auxiliary steering gear each, or two or more identical actuating systems; and **automatic synchronisation of thruster rotation** unless each thruster withstands the others’ forces ([4.1.2]).
- **$> 2\,500\text{ kW}$ per unit:** alternative power supply **automatically within 45 s**, capacity $\geq 30\text{ min}$ for ships $\geq 10\,000\text{ GT}$ and $\geq 10\text{ min}$ otherwise ([4.2.8]).
- **Controls** from the navigation bridge, machinery control station *and locally*, with a **thruster angle indicator independent of the control system** at each steering control station (Sec. 15 [2.4.1]).

- **Water-jets as steering means will be given special consideration** by the Society ([4.3.1]).

This applies to:

- Azimuth thrusters used for steering
- Podded propulsion units providing steering
- Water-jet systems providing steering
- Cycloidal (Voith-Schneider) propellers used for steering
- Any thruster configuration that replaces or supplements rudder steering

9.2 Additional Steering Gear Requirements

All applicable thruster drawings from Chapters 3-5 apply, **PLUS** the following steering-specific submissions:

9.2.1 Steering System Design

Table 9.1: Steering Gear - System Design

#	Drawing/Document	Category	Content Requirements	Reference
1	Steering System General Arrangement	FA	• Complete steering system layout • Main and auxiliary steering provisions • Control stations • Power supply arrangements • Emergency steering provisions	NR467 Sec.14, [1]
2	Steering Gear Capacity Calculations	FA	• Maximum steering torque • Time to move from 35° port to 35° starboard • Time to move from 35° to 30° on opposite side • Compliance with SOLAS requirements • Redundancy analysis	NR467 Sec.14, [2]; SOLAS Ch.II-1, Reg.29
Continued on next page				

Table 9.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
3	Steering Control System	FA	• Control system architecture • Main control station (bridge) • Auxiliary control provisions • Emergency control • Autopilot integration • Manual override provisions	NR467 Sec.14, [3]

9.2.2 Redundancy & Reliability

Table 9.2: Steering Gear - Redundancy Requirements

#	Drawing/Document	Category	Content Requirements	Reference
4	Main and Auxiliary Steering Systems	FA	• Main steering arrangement • Auxiliary steering arrangement • Independence of systems • Changeover provisions • Simultaneous operation capability	NR467 Sec.14, [4]; SOLAS Ch.II-1, Reg.29
5	Failure Mode and Effects Analysis (FMEA)	FA	• All potential failure modes • Effects on steering capability • Single failure tolerance • Redundancy verification • Mitigation measures	NR467 Sec.14, [5]
6	Reliability Analysis	FA	• System reliability calculations • Component failure rates • Mean time between failures (MTBF) • Availability analysis • Compliance with steering gear standards	NR467 Sec.14, [5]

9.2.3 Power Supply

Table 9.3: Steering Gear - Power Supply

#	Drawing/Document	Category	Content Requirements	Reference
7	Steering Gear Power Supply	FA	• Main power supply arrangement • Emergency power supply • Independent power sources for main/auxiliary • Switchboard arrangements • Circuit protection	NR467 Sec.14, [6]; SOLAS Ch.II-1, Reg.29
8	Power Supply Calculations	FA	• Load analysis • Cable sizing • Protection device coordination • Voltage drop calculations • Short circuit analysis	NR467 Sec.14, [6]; IEC 60092-352

9.2.4 Control Stations

Table 9.4: Steering Gear - Control Stations

#	Drawing/Document	Category	Content Requirements	Reference
9	Bridge Control Console	FA	• Main steering control layout • Steering wheel or joystick • Rudder angle indicator (or thruster angle) • Rate of turn indicator • Alarm and monitoring displays • Autopilot controls	NR467 Sec.14, [7]
10	Emergency Steering Station	FA	• Emergency control location • Control equipment • Communication with bridge • Indication and monitoring • Access provisions	NR467 Sec.14, [8]; SOLAS Ch.II-1, Reg.29
11	Local Control Station	FA	• Local control panel (if applicable) • Control mode selection • Indication and monitoring • Interlock with other control stations	NR467 Sec.14, [9]

9.2.5 Alarms and Monitoring

Table 9.5: Steering Gear - Alarms & Monitoring

#	Drawing/Document	Category	Content Requirements	Reference
12	Alarm System Design	FA	• Power supply failure alarms • Control system failure alarms • Actuator failure alarms • Position deviation alarms • Alarm display on bridge	NR467 Sec.14, [10]
13	Monitoring System	FA	• Position monitoring • Rate monitoring • Torque/thrust monitoring • Temperature monitoring • Data logging	NR467 Sec.14, [10]

9.2.6 Hydraulic Systems (if applicable)

Table 9.6: Steering Gear - Hydraulic System

#	Drawing/Document	Category	Content Requirements	Reference
14	Steering Gear Hydraulic System	FA	• Main hydraulic system • Auxiliary hydraulic system • Independence of systems • Accumulator provisions • See Chapter 11 for detailed requirements	NR467 Sec.14, [11]; Ch.11

9.2.7 Testing & Commissioning

Table 9.7: Steering Gear - Testing Requirements

#	Drawing/Document	Category	Content Requirements	Reference
15	Steering Gear Testing Procedures	FA	• Factory acceptance tests • Installation tests • Sea trial procedures • Performance verification • Redundancy testing • Emergency steering tests	NR467 Sec.14, [12]; SOLAS Ch.II-1, Reg.29
16	Steering Gear Test Records	FI	• FAT reports • Installation test reports • Sea trial results • Performance verification • As-built documentation	NR467 Sec.14, [12]

9.2.8 Operating Manuals

Table 9.8: Steering Gear - Operating Manuals

#	Drawing/Document	Category	Content Requirements	Reference
17	Steering Gear Operating Manual	FI	• Normal operating procedures • Changeover procedures (main/auxiliary) • Emergency procedures • Troubleshooting guide • Safety precautions	NR467 Sec.14, [13]
18	Maintenance Manual	FI	• Maintenance schedule • Inspection procedures • Lubrication requirements • Spare parts list • Overhaul procedures	NR467 Sec.14, [13]

9.2.9 SOLAS Compliance Documentation

Table 9.9: Steering Gear - SOLAS Compliance

#	Drawing/Document	Category	Content Requirements	Reference
19	SOLAS Compliance Statement	FA	• Demonstration of compliance with SOLAS Ch.II-1, Reg.29 • Steering gear capacity verification • Redundancy verification • Emergency steering capability • Testing and commissioning compliance	NR467 Sec.14, [14]; SOLAS Ch.II-1, Reg.29

Chapter 10

Thrusters as Part of Main Shafting

10.1 Applicability

When thruster shafts (propeller shafts, intermediate shafts, vertical shafts in azimuth thrusters, or water-jet drive shafts) are considered part of the main propulsion shafting system, they **MUST** comply with shafting requirements in NR467 Section 7.

This typically applies to:

- Azimuth thruster vertical and horizontal shafts
- Podded propulsion shafts
- Water-jet drive shafts
- Tunnel thruster propeller shafts (for high-power units)
- Any shaft transmitting significant propulsion power

10.2 Overlap with Thruster Requirements

NOTE: Many shaft-related drawings and calculations are already covered in Chapters 3-5. This chapter identifies **ADDITIONAL** requirements or **ENHANCED** detail levels required when shafts are classified as "main shafting."

10.2.1 Shaft Design Drawings

Table 10.1: Main Shafting - Design Drawings

#	Drawing/Document	Category	Content Requirements	Reference
1	Shaft Line General Arrangement	FA	• Complete shaft line layout • All shaft sections identified • Bearing locations • Coupling locations • Seal locations • Coordinate system	NR467 Sec.7, [1]
2	Shaft Detail Drawings	FA	• Dimensional drawings of each shaft section • Diameter changes and transitions • Keyway details • Thread details • Surface finish requirements • Material specifications • Heat treatment requirements	NR467 Sec.7, [2]
3	Shaft End Connections	FA	• Flange designs • Coupling designs • Bolt specifications • Fit tolerances • Assembly procedures	NR467 Sec.7, [3]

10.2.2 Shaft Strength Calculations

Table 10.2: Main Shafting - Strength Calculations

#	Drawing/Document	Category	Content Requirements	Reference
4	Shaft Diameter Calculations	FA	• Minimum diameter calculations per BV formula • Torsional strength verification • Bending strength verification • Combined stress analysis • Safety factors applied	NR467 Sec.7, [4]
5	Shaft Stress Analysis	FA	• Detailed stress calculations at critical sections • Stress concentration factors • Keyway effects • Thread effects • FEA results (if applicable)	NR467 Sec.7, [4]
Continued on next page				

Table 10.2 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
6	Fatigue Analysis	FA	• Fatigue life calculations • S-N curves for materials • Stress range analysis • Cumulative damage assessment • Compliance with fatigue criteria	NR467 Sec.7, [5]
7	Deflection Calculations	FA	• Shaft deflection under load • Bearing alignment effects • Seal clearance verification • Acceptable deflection limits	NR467 Sec.7, [6]

10.2.3 Critical Speed Analysis

Table 10.3: Main Shafting - Critical Speed Analysis

#	Drawing/Document	Category	Content Requirements	Reference
8	Torsional Vibration Analysis	FA	• Natural frequencies calculation • Forced vibration analysis • Resonance identification • Barred speed ranges • Damping analysis • Compliance with acceptable vibration levels	NR467 Sec.7, [7]
9	Lateral Vibration Analysis	FA	• Critical speed calculations • Campbell diagram • Operating speed verification • Margin to critical speeds • Bearing stiffness effects	NR467 Sec.7, [7]
10	Whirling Analysis	FA	• Whirling speed calculations • Forward and backward whirl • Operating range verification • Stability analysis	NR467 Sec.7, [7]

10.2.4 Bearing Design

Table 10.4: Main Shafting - Bearing Design

#	Drawing/Document	Category	Content Requirements	Reference
11	Bearing Arrangement Drawing	FA	• Bearing types and locations • Bearing housing design • Lubrication system • Seal arrangements • Cooling provisions • Mounting details	NR467 Sec.7, [8]
12	Bearing Load Calculations	FA	• Radial loads • Axial loads (thrust bearings) • Moment loads • Load distribution among bearings • Dynamic load factors	NR467 Sec.7, [8]
13	Bearing Life Calculations	FA	• L10 life calculations • Operating conditions • Load and speed effects • Lubrication effects • Minimum acceptable life	NR467 Sec.7, [8]
14	Bearing Lubrication System	FA	• Lubrication method (oil bath, forced circulation, grease) • Oil type and properties • Oil flow rate (forced systems) • Cooling arrangements • Filtration system • Oil level and temperature monitoring	NR467 Sec.7, [9]

10.2.5 Couplings

Table 10.5: Main Shafting - Couplings

#	Drawing/Document	Category	Content Requirements	Reference
15	Coupling Design Drawing	FA	• Coupling type (rigid, flexible) • Dimensions and material • Bolt specifications • Fit tolerances • Assembly procedure	NR467 Sec.7, [10]
Continued on next page				

Table 10.5 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
16	Coupling Strength Calculations	FA	• Torque capacity • Bolt strength verification • Flange strength • Fit stress analysis • Fatigue assessment	NR467 Sec.7, [10]
17	Flexible Coupling Characteristics	FA	• Torsional stiffness • Damping properties • Misalignment capacity • Effect on torsional vibration	NR467 Sec.7, [10]

10.2.6 Seals

Table 10.6: Main Shafting - Seals

#	Drawing/Document	Category	Content Requirements	Reference
18	Seal System Design	FA	• Seal type and material • Seal arrangement • Pressure rating • Leakage monitoring provisions • Emergency sealing (if applicable)	NR467 Sec.7, [11]
19	Seal Performance Data	FA	• Operating pressure and temperature • Shaft speed limitations • Expected leakage rates • Service life • Maintenance requirements	NR467 Sec.7, [11]

10.2.7 Alignment

Table 10.7: Main Shafting - Alignment

#	Drawing/Document	Category	Content Requirements	Reference
20	Shaft Alignment Calculations	FA	• Alignment procedure • Bearing reactions • Shaft sag calculations • Thermal growth considerations • Acceptable alignment tolerances	NR467 Sec.7, [12]
21	Alignment Procedure	FI	• Step-by-step alignment procedure • Measurement methods • Adjustment provisions • Verification methods • Documentation requirements	NR467 Sec.7, [12]

10.2.8 Material Specifications

Table 10.8: Main Shafting - Material Specifications

#	Drawing/Document	Category	Content Requirements	Reference
22	Shaft Material Specifications	FA	• Material grades (steel, stainless steel, etc.) • Mechanical properties • Heat treatment requirements • Surface treatment (if applicable) • Corrosion protection • Compliance with BV approved materials	NR467 Sec.7, [13]; Part B, Ch.3
23	Material Certificates	FA	• Material test certificates (during construction) • Chemical composition • Mechanical properties • Heat treatment records • Traceability	NR467 Part B, Ch.3

10.2.9 Non-Destructive Testing (NDT)

Table 10.9: Main Shafting - NDT Requirements

#	Drawing/Document	Category	Content Requirements	Reference
24	NDT Plan	FA	• NDT methods (UT, MT, PT, RT) • Extent of testing • Acceptance criteria • Inspector qualifications • Testing stages	NR467 Sec.7, [14]; Part B, Ch.4
25	NDT Reports	FI	• NDT test reports (during construction) • Test results • Defect identification and disposition • Inspector certification	NR467 Part B, Ch.4

10.2.10 Testing & Commissioning

Table 10.10: Main Shafting - Testing & Commissioning

#	Drawing/Document	Category	Content Requirements	Reference
26	Shaft Testing Procedures	FA	• Dimensional verification • Material testing • NDT • Balance testing (if applicable) • Assembly testing • Acceptance criteria	NR467 Sec.7, [15]
27	Vibration Monitoring Plan	FA	• Vibration measurement locations • Measurement methods • Acceptance criteria • Monitoring during sea trials • Long-term monitoring provisions	NR467 Sec.7, [16]
28	Test Reports	FI	• Factory test reports • Installation test reports • Sea trial vibration measurements • Alignment verification • As-built documentation	NR467 Sec.7, [15]

10.3 MON-SHAFT & Shaft-Line Condition-Monitoring Notations

Where the thruster forms part of the propulsion shaft line (azimuth, podded, water-jet or high-power tunnel units with oil-lubricated stern-tube/lower-unit bearings), the owner may request the **MON-SHAFT** additional class notation. MON-SHAFT allows the classical periodic *tailshaft withdrawal / complete shaft survey* to be replaced by a survey regime based on **continuous monitoring and condition assessment** of the oil-lubricated shaft/bearing/seal arrangement. It therefore *adds documentation* to the shaft-line package and interacts directly with the thruster's lower-unit bearing, seal and oil systems.

Table 10.11: MON-SHAFT — Additional Shaft-Line Monitoring Submissions

#	Drawing/Document	Category	Content Requirements	Reference
1	Oil-Lubricated Bearing & Seal Arrangement	FA	• Approved stern-tube / lower-unit oil-lubricated bearing & seal design • Wear-down measurement provisions • Compatibility with monitoring notation	NR467 Sec.7; Pt E (MON-SHAFT)
2	Lube-Oil Monitoring & Analysis Programme	FA	• Oil temperature (and pressure where applicable) monitoring • Periodic oil-analysis programme (contamination, wear metals, water) • Sampling points and acceptance/alert criteria	NR467 Pt E (MON-SHAFT)
3	Bearing Wear-down / Sensor Instrumentation	FA	• Bearing temperature sensors and locations • Wear-down/liner-wear measurement method & trending • Alarm set-points and data logging	NR467 Pt E; Pt C, Ch.3
4	Survey Arrangement & Data-Management Plan	FI	• Monitoring-based survey scheme replacing shaft withdrawal • Record keeping, trend review, thresholds for action • Interface with planned maintenance / class survey	NR467 Pt A (surveys); Pt E

Surveyor note (MON-SHAFT): MON-SHAFT is only compatible with **oil-lubricated** shaft/bearing arrangements of an approved type. For azimuth and podded units the “shaft line” extends into the lower gear/pod; confirm the seal, bearing and oil-monitoring documentation of

the thruster lower unit (Chapters 4) is consistent with the MON-SHAFT scheme, and that the oil-analysis and wear-down-trending programme, sensors and acceptance criteria are all submitted. Related shaft-line notations (e.g. oil-lubricated stern-tube monitoring / **TIGHT**-type arrangements) follow the same principle — verify the exact notation and its rule article in the current NR467 Part E.

Chapter 11

Hydraulic Systems & Pitch Control Mechanisms

11.1 Applicability

Hydraulic systems are used in thrusters for:

- Hydraulic motor drive (alternative to electric motor)
- Retractable thruster lifting/lowering mechanism
- Controllable pitch propeller pitch control
- Azimuth slewing drive (if hydraulic)
- Water-jet steering and reversing actuation
- Voith-Schneider control (steering) servo system

Rule basis: Thruster hydraulic systems are assessed under **NR467 Pt C, Ch 1** — the thruster/water-jet requirements of Sec. 15, the **pipng-system rules** (design pressure, materials, testing, relief), and, where the hydraulics actuate a *steering* function, the more stringent **steering-gear requirements of Sec. 14** (duplication, single-failure, power supply, response time). Recognised standard **ISO 4413** applies to general hydraulic-system design. Steering-critical hydraulics attract the redundancy and control provisions of Chapter 9; pitch-control hydraulics must define the *fail-safe* pitch position.

11.2 Hydraulic Power Unit (HPU)

Table 11.1: Hydraulic Systems - HPU

#	Drawing/Document	Category	Content Requirements	Reference
1	HPU General Arrangement	FA	• Complete HPU layout • Pump arrangement • Motor/drive arrangement • Reservoir design • Valve manifold • Accumulator location • Filtration system • Cooling system • Mounting arrangement	NR467 Sec.15
2	Hydraulic Pump Specification	FA	• Pump type (gear, piston, vane) • Displacement • Rated pressure and flow • Drive motor specifications • Efficiency curves • Manufacturer and model	NR467 Sec.15
3	Hydraulic Reservoir Design	FA	• Reservoir capacity • Material and construction • Baffle arrangement • Oil level indication • Breather and filler • Drain provisions • Heating elements (if applicable)	NR467 Sec.15
4	HPU Calculations	FA	• Flow requirements • Pressure requirements • Pump sizing verification • Reservoir sizing • Heat generation and cooling capacity • Accumulator sizing (if applicable)	NR467 Sec.15

11.3 Hydraulic Circuit Design

Table 11.2: Hydraulic Systems - Circuit Design

#	Drawing/Document	Category	Content Requirements	Reference
5	Hydraulic System Schematic	FA	• Complete hydraulic circuit • All components identified with symbols • Pressure ratings • Flow ratings • Control logic • Safety devices • Monitoring points	NR467 Sec.15
6	Piping and Instrumentation Diagram (P&ID)	FA	• Detailed piping layout • Pipe sizes and materials • Valve locations and specifications • Instrument locations • Pressure and temperature monitoring • Drain and vent points	NR467 Sec.15
7	Hydraulic Piping Arrangement	FA	• 3D piping layout (or isometric) • Pipe routing • Pipe supports • Flexibility analysis • Penetrations through bulk-heads • Accessibility for maintenance	NR467 Sec.15

11.4 Hydraulic Components

Table 11.3: Hydraulic Systems - Components

#	Drawing/Document	Category	Content Requirements	Reference
8	Hydraulic Motor Specification	FA	• Motor type and displacement • Rated pressure and speed • Torque and power curves • Efficiency • Mounting details • Manufacturer and model	NR467 Sec.15
Continued on next page				

Table 11.3 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
9	Hydraulic Cylinder Specification	FA	• Cylinder bore and stroke • Operating pressure • Rod diameter and material • Seal specifications • Mounting arrangement • Position sensors • Manufacturer and model	NR467 Sec.15
10	Control Valve Specifications	FA	• Valve types (directional, pressure, flow) • Valve ratings • Actuation method (solenoid, manual, pilot) • Response time • Manufacturer and model	NR467 Sec.15
11	Pressure Relief Valve	FA	• Set pressure • Flow capacity • Type (direct-acting, pilot-operated) • Mounting location • Manufacturer and model	NR467 Sec.15
12	Accumulator Specification	FA	• Accumulator type (bladder, piston, diaphragm) • Volume • Pre-charge pressure • Operating pressure range • Gas type (nitrogen) • Mounting arrangement • Safety provisions	NR467 Sec.15
13	Accumulator Sizing Calculations	FA	• Volume requirements • Pre-charge pressure calculation • Energy storage capacity • Response time analysis	NR467 Sec.15

11.5 Filtration System

Table 11.4: Hydraulic Systems - Filtration

#	Drawing/Document	Category	Content Requirements	Reference
14	Filtration System Design	FA	• Filter types and locations • Filtration rating (micron) • Flow capacity • Pressure drop • Filter change indicators • Bypass provisions	NR467 Sec.15
15	Filter Specifications	FA	• Filter element type • Filtration rating • Collapse pressure • Manufacturer and model • Replacement schedule	NR467 Sec.15

11.6 Cooling System

Table 11.5: Hydraulic Systems - Cooling

#	Drawing/Document	Category	Content Requirements	Reference
16	Hydraulic Oil Cooling System	FA	• Cooling method (air, water) • Heat exchanger specifications • Cooling capacity • Temperature control • Monitoring provisions	NR467 Sec.15
17	Thermal Analysis	FA	• Heat generation calculations • Heat dissipation calculations • Temperature rise predictions • Cooling capacity verification • Operating temperature range	NR467 Sec.15

11.7 Hydraulic Fluid

Table 11.6: Hydraulic Systems - Fluid Specification

#	Drawing/Document	Category	Content Requirements	Reference
18	Hydraulic Fluid Specification	FA	- Fluid type (mineral oil, synthetic, biodegradable) - Viscosity grade - Operating temperature range - Fire resistance properties (if applicable) - Compatibility with system materials - Approved manufacturer and grade	NR467 Sec.15

11.8 Piping Design

Table 11.7: Hydraulic Systems - Piping

#	Drawing/Document	Category	Content Requirements	Reference
19	Pipe Sizing Calculations	FA	- Flow velocity calculations - Pressure drop calculations - Pipe diameter selection - Compliance with velocity limits	NR467 Sec.15
20	Pipe Material Specifications	FA	- Pipe material (steel, stainless steel, copper-nickel) - Wall thickness - Pressure rating - Corrosion protection - Compliance with BV approved materials	NR467 Sec.15; Part B, Ch.3
21	Pipe Support Design	FA	- Support locations and types - Support spacing - Vibration isolation - Thermal expansion accommodation - Load calculations	NR467 Sec.15
22	Piping Stress Analysis	FA	- Pressure stress - Thermal stress - Vibration analysis - Support reaction forces - Compliance with stress limits	NR467 Sec.15

11.9 Hose Assemblies

Table 11.8: Hydraulic Systems - Hoses

#	Drawing/Document	Category	Content Requirements	Reference
23	Hydraulic Hose Specifications	FA	• Hose type and construction • Pressure rating • Temperature rating • End fittings • Length and routing • Fire resistance (if required) • Manufacturer and model	NR467 Sec.15
24	Hose Installation Drawing	FA	• Hose routing • Bend radius verification • Support and protection • Accessibility for inspection/replacement	NR467 Sec.15

11.10 Safety Devices

Table 11.9: Hydraulic Systems - Safety Devices

#	Drawing/Document	Category	Content Requirements	Reference
25	Pressure Relief System	FA	• Relief valve locations • Set pressures • Discharge routing • Capacity verification	NR467 Sec.15
26	Emergency Shutdown System	FA	• Emergency stop provisions • Valve actuation • System depressurization • Safe state definition	NR467 Sec.15
27	Leak Detection System	FA	• Leak detection method • Alarm provisions • Automatic shutdown (if required)	NR467 Sec.15

11.11 Control System

Table 11.10: Hydraulic Systems - Control

#	Drawing/Document	Category	Content Requirements	Reference
28	Hydraulic Control System	FA	• Control architecture • Manual/automatic control modes • Valve actuation signals • Feedback systems • Safety interlocks	NR467 Sec.15; Part C, Ch.4
29	Control Panel Layout	FA	• Panel arrangement • Control devices • Indication and monitoring • Alarm displays • Emergency stop	NR467 Sec.15

11.12 Monitoring & Instrumentation

Table 11.11: Hydraulic Systems - Monitoring

#	Drawing/Document	Category	Content Requirements	Reference
30	Instrumentation List	FA	• Pressure gauges/transmitters • Temperature sensors • Flow meters (if applicable) • Level indicators • Position sensors • Filter condition indicators • Specifications for each instrument	NR467 Sec.15
31	Alarm and Monitoring System	FA	• Parameters monitored • Alarm setpoints • Alarm display location • Data logging (if applicable)	NR467 Sec.15

11.13 Controllable Pitch Mechanism (CP Propellers)

Table 11.12: Hydraulic Systems - CP Mechanism

#	Drawing/Document	Category	Content Requirements	Reference
32	CP Mechanism Assembly	FA	• Hub internal mechanism • Pitch control rod/linkage • Hydraulic cylinder (in hub or external) • Blade retention system • Bearing arrangement • Seal design	NR467 Sec.15, [3.4]
33	CP Mechanism Calculations	FA	• Pitch control forces • Hydraulic cylinder sizing • Linkage strength • Blade retention strength • Seal pressure rating	NR467 Sec.15, [3.4]
34	Pitch Control Hydraulic System	FA	• Hydraulic supply to hub • Rotary joint/slip ring design • Control valve system • Pitch feedback system • Emergency pitch setting	NR467 Sec.15, [3.4]
35	Pitch Feedback System	FA	• Position sensor type and location • Signal transmission (rotary joint) • Accuracy and resolution • Redundancy (if required)	NR467 Sec.15, [3.4]

11.14 Testing & Commissioning

Table 11.13: Hydraulic Systems - Testing

#	Drawing/Document	Category	Content Requirements	Reference
36	Hydraulic System Test Procedures	FA	• Pressure testing • Functional testing • Leak testing • Performance verification • Safety device testing • Acceptance criteria	NR467 Sec.15
Continued on next page				

Table 11.13 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
37	Hydraulic System Test Reports	FI	• Pressure test results • Functional test results • Leak test results • Performance data • As-built documentation	NR467 Sec.15

11.15 Operating & Maintenance Manuals

Table 11.14: Hydraulic Systems - Manuals

#	Drawing/Document	Category	Content Requirements	Reference
38	Hydraulic System Operating Manual	FI	• System description • Operating procedures • Normal operation • Emergency procedures • Troubleshooting guide • Safety precautions	NR467 Sec.15
39	Hydraulic System Maintenance Manual	FI	• Maintenance schedule • Inspection procedures • Fluid sampling and analysis • Filter replacement • Component overhaul • Spare parts list	NR467 Sec.15

Chapter 12

Dynamic Positioning (DP) Thrusters - Additional Requirements

12.1 Applicability

NR467 Sec. 15 [1.1.1]: “For azimuth thrusters intended for dynamic positioning, the additional requirements in **Pt F, Ch. 11, Sec. 5** are to be complied with.” Those requirements are *in addition to* the full type package for each thruster used in the DP system.

The authoritative DP submission list is Table 1.14 (NR467 Pt F, Ch. 11, Sec. 5, Tab. 1 — 22 rows, reproduced verbatim in Section 1.12). This chapter gives the surveyor’s reading of it. DYNAPOS [1.1.3] cross-refers back to Pt C, Ch. 1, Sec. 2 (engine-driven thrusters), **Pt C, Ch. 1, Sec. 15** (azimuthal and transverse thrusters) and Pt F, Chapter 3 (automation).

12.1.1 Notation symbols

The additional class notation is **DYNAPOS**, completed by symbols describing the capability of the installation: **AM**, **AT**, **AM/AT** and **SAM**, with the redundancy-qualified variants **AM/AT R** and **AM/AT RS** (NR467 Pt A, Ch. 1, Sec. 2, [6.21.5]; Pt F, Ch. 11, Sec. 5). **DYNAPOS AM/AT** is equivalent to IMO equipment class 1 ([3.2.1]). An environmental station keeping index (**ESKI**) may be associated with the notation ([9.1.1]).

The FMEA is conditional — do not demand it as a matter of course. Table 1 item 17 (Failure Mode and Effect Analysis) is annotated in the rule “**For symbols R and RS only**”, and [2.4.1] confirms: “For installation intended to be assigned the notation **DYNAPOS AM/AT R** and **DYNAPOS AM/AT RS** an FMEA is to be carried out.”

A plain DYNAPOS AM, AT or AM/AT installation does not attract an FMEA under Pt F, Ch. 11, Sec. 5. Earlier editions of this catalog stated unconditionally that the FMEA is the cornerstone DP document and that an inadequate FMEA is grounds for rejection. That is correct *for R and RS installations* and wrong for the rest. Check the requested notation string first.

Note the related worst-case-failure provision: for **AM/AT R** or **AM/AT RS**, the DP vessel is to be operated so that the worst-case failure determined in [2.4.1] can occur at any time without breaching the acceptable excursion criteria ([3.1.1]). For **AM/AT**, the power system need not be redundant ([4.2.4]).

12.2 Thruster-specific rows in the DYNAPOS table

Three rows bear directly on the thruster package and are easily overlooked:

- **Item 11 (FA)** — for approval of propulsion based on rotary azimuth thrusters: *layout drawings of thrust units, thrust shafts and blocks; arrangement of hull passages; thrust curves of each propulsion unit.*
- **Item 18 (FA)** — *study of possible interaction between thrusters.*
- **Item 3 (FA)** — *capability plots, covering the conditions in the specification, the case with all thrusters running and selected in DP, and the worst case failure.*

12.3 Reference

NR467 Part F, Chapter 11, Section 5 (DYNAPOS) is the governing text; DP control system type testing follows **Pt C, Ch. 3, Sec. 6** (item 10). See also IMO MSC.1/Circ.1580 and IMCA guidance, and **NR632** for hardware-in-the-loop testing of DP control systems.

Chapter 13

Ice Class & Cold-Climate (Winterisation) Requirements

13.1 Applicability — and the scope split that matters most

Read this before demanding anything in this chapter. The ice scope depends entirely on the thruster’s **duty**. NR467 Sec. 15 [1.1.1], verbatim:

“Thrusters intended for **propulsion and steering** of ships with an ICE CLASS notation are to comply with the additional requirements of Part F, Chapter 8. **Transverse thrusters intended for manoeuvring** of ships with an ICE CLASS notation **are required to comply with additional requirement Pt F, Ch. 8, Sec. 3, [3.5.1] only.**”

And [3.5.1] in its entirety reads: “*Tunnels of transverse thrusters are to be fitted with grids for protection against ice impacts.*”

So for a manoeuvring bow or stern tunnel thruster on an ice-class ship, the entire ice requirement is a protective grid on the tunnel. The ice-load strengthening package in this chapter applies to *propulsion and steering* thrusters. Applying it to a manoeuvring tunnel unit is a substantial over-demand and a common review error.

The notations, per NR467 Pt F, Ch. 8, Sec. 1, [1.1.1], are: **ICE CLASS IA SUPER**, **ICE CLASS IA**, **ICE CLASS IB**, **ICE CLASS IC**, **ICE CLASS ID**, **YOUNG ICE 1** and **YOUNG ICE 2**. Per [1.1.2] the strengthening requirements — *excepting* ICE CLASS ID, YOUNG ICE 1 and YOUNG ICE 2 — are equivalent to the **Finnish-Swedish Ice Class Rules 2019, as amended**, for ships trading in the Baltic in winter or equivalent ice conditions.

Polar service. Thrusters intended for propulsion and steering of ships with an additional class notation **POLAR CLASS** or a service notation **icebreaker** comply with NR527 (Sec. 15 [1.1.1]). Where the vessel has an **azimuth** propulsion system, the controlling document is NR584 — see Section 1.14. Machinery ice loads follow **IACS UR I3**, whose propulsion machinery type factor is $k_3 = 1,0$ for fixed and 1,2 for azimuthing propulsors.

Cold-climate service without ice loads is covered by winterisation notations such as **COLD**, addressing design ambient temperatures rather than ice impact (NR467 Pt F, Ch. 8, Sec. 4).

13.2 Ice-Strengthening Submissions

Applies to thrusters intended for *propulsion and steering* only — see the scope box above. The authoritative sub-article map is Table 1.15; the table below is the supporting

engineering content. For an azimuth unit, note especially Pt F, Ch. 8, Sec. 3, [1.11] — thruster body ice loads, global vibration, **steering gear design torque in ice** and protection of the steering gear against excessive torque.

Table 13.1: Ice Class — Additional Thruster Requirements

#	Drawing/Document	Category	Content Requirements	Reference
1	Ice Class Notation Basis & Design Conditions	FA	- Declared notation (IA Super/IA/... or PC1–PC7) - Design ice thickness / ice-load basis - Applicable rule edition and lower design temperature	NR467 Part F; NR527
2	Propeller Ice-Load Strength Calculation	FA	- Ice interaction loads (milling, impact) on blades - Blade root and blade tip strength vs. ice bending - Blade material and thickness increase - Open vs. ducted / nozzle ice loads	NR467 Part E; IACS UR I3
3	Shaft Line & Ice Torque Verification	FA	- Maximum ice-induced torque and thrust - Propeller shaft, intermediate shaft, couplings - Overload/peak-torque capacity of gears - Blade-loss / one-blade-milling scenarios	NR467 Part E; Sec.7; IACS UR I3
4	Gear & Azimuth/Nozzle Ice-Load Substantiation	FA	- Bevel/reduction gears vs. ice peak torque - Azimuth slewing bearing/gear vs. ice-block impact on lower unit - Nozzle/steering-grid ice-impact strength	NR467 Part E; Sec.15
5	Thruster Body / Lower Unit Ice Protection	FA	- Lower gearbox/pod shape vs. ice interaction - Ice knife / deflector where required - Local structural strengthening	NR467 Part E
Continued on next page				

Table 13.1 – continued from previous page

#	Drawing/Document	Category	Content Requirements	Reference
6	Tunnel / Inlet Ice & Grid Arrangement	FA	• Tunnel thruster grid ice/debris protection • De-icing / anti-blocking provisions • Ice-load on grid bars	NR467 Part E; Sec.15
7	Winterisation of Systems (Hydraulic, Cooling, Seals)	FA	• Low-temperature grades of oils/greases/seals • Heating/insulation of exposed hydraulics & sensors • Sea-chest/cooling anti-icing (recirculation) • Materials for low-temperature toughness (Charpy)	NR467 Part E (COLD); Part B, Ch.3
8	Materials Low-Temperature Certification	FA	• Impact testing at design temperature • Grade selection for exposed & load-bearing parts	NR467 Part B, Ch.3; Part E
9	Retraction/Deployment in Ice (if retractable)	FI	• Operating limits and procedures in ice • Protection of opening/seals from ice ingress	NR467 Part E

Surveyor note (ice): The governing case for ice-classed propulsion is usually the **ice peak torque / blade-milling load** propagating through propeller → shaft → gear → prime mover, *not* the open-water rating. Verify the ice-load calculation explicitly and check that it is carried consistently into the shaft (Chapter 10) and gear strength checks. For azimuth/podded units in ice, confirm lower-unit and slewing-gear substantiation against ice-block impact — a frequent omission. Match the rule edition to the declared notation.

Chapter 14

Additional BV Class Notations Affecting Thruster Packages

14.1 Purpose

Beyond ice class, several BV **additional class notations** and **service notations** impose extra requirements on the thruster package. This chapter summarises the ones most often encountered so the surveyor can confirm the client has submitted the corresponding documents. Requirements here *add to* the type package; they do not replace it.

14.2 Notation Impact Matrix

Table 14.1: Additional Notations — Impact on Thruster Submissions

Notation (family)	Concern	Additional thruster submissions	Reference
DYNAPOS (AM / AT / AM-AT / AM-AT R / AM-AT RS / SAM); DP-1/2/3	Dynamic positioning	Full DP package — see Chapter 12: FMEA, redundancy, power segregation, capability plot, thruster allocation, proving trials.	NR467 Pt F, Ch.11 (DYNAPOS)
AUT-UMS, AUT-CCS, AUT-PORT	Automated / unattended machinery	Extended alarm & monitoring of thruster (bearing temps, oil, vibration), remote & safe control, fail-safe on power/control loss, integration with alarm system.	NR467 Pt C, Ch.3; Pt F, Ch.3 (AUT)
COMF-NOISE / COMF-VIB (Comfort)	Noise & vibration on board	Structure-borne & airborne noise prediction, vibration levels at thruster excitation (blade-rate), resilient mounting, cavitation-noise control, source data.	NR467 Pt F, Ch.6 (COMF)
Continued on next page			

Table 14.1 – continued from previous page

Notation (family)	Concern	Additional thruster submissions	Reference
CLEANSHIP / environmental	Pollution prevention	Environmentally Acceptable Lubricants (EAL) for stern-tube/thruster seals & hydraulics, leakage monitoring, oil-to-sea interface documentation.	NR467 Pt F, Ch.9 (environmental); VGP/EAL
Redundant Propulsion (AVM-DPS)	Propulsion availability	Independence/segregation of propulsion trains, single-failure analysis, take-me-home capability, redundant thruster arrangement. Related: HEADING CONTROL-DS / HEADING CONTROL-IS for heading-keeping with redundant propulsion/steering.	NR467 Pt F, Ch.2 (AVM-DPS); Ch.11 (HEADING CONTROL)
MON-SHAFT (shaft-line monitoring)	Shaft survey by condition monitoring	Oil-lubricated bearing/seal of approved type; lube-oil monitoring & analysis programme; bearing temperature & wear-down instrumentation; monitoring-based survey scheme (see Section 10.3).	NR467 Pt F, Ch.5; Sec.7
TUG / ESCORT TUG / AHT (service)	High steering & towing loads	Escort/indirect-towing loads on azimuth & steering system, structural substantiation of unit & foundation, stability interaction, uprated steering duty.	NR467 Pt E, Ch.1 (Tugs) / Ch.2 (AHT)
SPS (Special Purpose Ship)	Personnel / mission systems	Higher survivability & redundancy expectations may extend to propulsion/steering arrangement.	NR467 Pt D/E
Electric hybrid / battery	Hybrid power	Documentation of every driving path (PTI/PTO), energy-storage interface, mode changeover, converter data.	NR467 Pt F, Ch.14 (Electric Propulsion)

Surveyor note (notations): Always cross-check the vessel's *requested class notation string* against the thruster package. A DP or comfort notation in particular pulls in substantial extra documentation (FMEA/proving trials; noise & vibration studies) that clients frequently omit from the initial thruster submission. Where a notation adds a steering-critical duty (escort tug), confirm the steering-gear package (Chapter 9) reflects the *uprated* loads, not just the open-water condition.

Chapter 15

Ship-Type & Service-Notation Specific Requirements

15.1 Purpose

The base thruster package (Chapters 2–7) and the prime-mover, ice and notation overlays (Chapters 8, 13, 14) define *what a thruster needs*. This chapter answers the complementary question the surveyor must also ask: **“given the ship type / service notation, what *extra* do I expect in the thruster package, and which *additional ship rules* apply?”**

BV assigns each ship a **service notation** (NR467 Part D) — e.g. tug, supply, oil tanker, passenger ship — plus **additional class notations** (Part E). Together these pull in requirements beyond NR467 Sec.15, and sometimes bring in *separate rule sets* (offshore units, naval ships, high-speed craft). The matrix below is the surveyor’s “what else to demand” guide; the subsequent sections expand the cases with the most rule impact.

15.2 Ship-Type → Thruster Expectation Matrix

Table 15.1: Ship Type / Service — Typical Arrangement & Additional Requirements

Ship type / service	Typical thruster arrangement	Additional requirements & notations to expect	Reference
Harbour / escort tug	Azimuth Z-drive or VSP (main propulsion + steering)	Escort/indirect-towing steering & structural loads; uprated slewing duty; steering redundancy; stability interaction; full steering-gear package.	NR467 Pt E, Ch.1 (tug/escort tug); Sec.14
Offshore supply / PSV / AHTS	Bow tunnel(s) + retractable + stern azimuth; DP	DYNAPOS/DP (usually DP-2): FMEA, redundancy, power segregation, capability plot; FiFi where fitted; station-keeping.	NR467 Pt F, Ch.11 (DYNAPOS); Pt E, Ch.2–3

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Table 15.1 – continued from previous page

Ship type / service	Typical thruster arrangement	Additional requirements & notations to expect	Reference
Dredger (TSHD / CSD)	Bow tunnels; azimuth; sometimes DP	DREDGING service; DP for some; robust manoeuvring; abrasion/duty considerations.	NR467 Pt D, Ch.13
Passenger ship / RoPax / ferry	Bow (+ stern) tunnels; azimuth/pods on some	SOLAS steering (II-1/29); Safe Return to Port survivability & redundancy for ships ≥ 120 m / 3 fire zones; COMF.	SOLAS II-1/II-2; Pt D, Ch.11–12
Cruise ship	Podded propulsors + multiple bow/stern tunnels	SRtP; COMF-NOISE/VIB; propulsion/steering redundancy; sometimes DP.	SOLAS II-2 Reg.21/22; Pt D, Ch.11
Double-ended ferry	Azimuth or VSP at both ends	Steering redundancy <i>at each end</i> ; control-station changeover; response time both directions.	Sec.14; SOLAS II-1/29
Cable / pipe layer, offshore construction	Retractable + azimuth + tunnels; DP	DYNAPOS DP-2/DP-3: FMEA, fire/flood separation (DP-3), redundancy, proving trials.	NR467 Pt F, Ch.11 (DYNAPOS); Pt E, Ch.6 / Ch.12
Diving support vessel (DSV)	Multiple azimuth/tunnel thrusters; DP-2/DP-3	DYNAPOS (usually DP-2/DP-3): FMEA, redundancy, power segregation, capability plot, proving trials; interfaces with the diving system itself (see NR610 for the diving system, distinct from the ship's NR467 class).	NR467 Pt E, Ch.7; Pt F, Ch.11 (DYNAPOS); NR610
Lifting unit / heavy-lift & crane vessel	Multiple azimuth/tunnel thrusters; often DP	DYNAPOS where fitted: FMEA, redundancy, capability plot; crane/lift loads interact with stability & thruster foundation loads.	NR467 Pt E, Ch.8; Pt F, Ch.11 (DYNAPOS) where DP fitted
Fire-fighting ship	Bow/stern tunnels or azimuth; FiFi pumps often share prime mover with thrusters	FiFi pump/thruster power-sharing and mode-changeover documentation; station-keeping during fire-fighting duty.	NR467 Pt E, Ch.4
Continued on next page			

Table 15.1 – continued from previous page

Ship type / service	Typical thruster arrangement	Additional requirements & notations to expect	Reference
Oil-recovery ship	Bow tunnel(s); sometimes azimuth	Station-keeping during recovery operations; hazardous-area considerations for recovered-oil handling spaces near the thruster room.	NR467 Pt E, Ch.5
Standby rescue vessel	Bow tunnel(s); sometimes azimuth	Manoeuvring/station-keeping close to platforms or vessels in distress; base Sec.15 package typically applies.	NR467 Pt E, Ch.10
Drillship / semi-sub (MODU)	Multiple azimuth thrusters; DP-3	DP-3 station keeping; offshore-unit rules; fire/flood zone separation; extensive FMEA.	NR467 Pt F, Ch.11 (DYNA-POS); NR445
Oil / chemical tanker	Bow thruster (manoeuvring)	Hazardous-area / explosion protection where equipment is in or adjacent to dangerous zones; thruster-room zoning; certified-safe (Ex) equipment.	SOLAS II-2; IEC 60079; Pt C; Pt D, Ch.7–8/10
Gas carrier (LNG/LPG)	Bow thruster; sometimes azimuth	As tanker <i>plus</i> IGC Code hazardous-zone requirements; gas-safe equipment.	IGC Code; IEC 60079; Pt D, Ch.9
Ice-class / polar trader	Ice-strengthened tunnel/azimuth	Ice-load strengthening & winterisation (see Chapter 13).	NR467 Pt F, Ch.8; NR527
Research / naval / patrol	Rim-driven / pods / azimuth	Low underwater-noise & signature; COMF/silent; naval ships follow dedicated naval rules.	Naval rules (e.g. NR483); Pt D, Ch.16 (patrol)
Large yacht	Bow/stern tunnels; pods	Comfort (noise/vibration); yacht-specific rules; retractable/stabiliser interfaces.	NR500 (yacht rules)
Container / bulk / general cargo	Bow tunnel thruster (manoeuvring)	Base Sec.15 package; typically electric drive; no special service overlay.	NR467 Sec.15; Pt C; Pt D, Ch.2/4

15.3 Passenger Ships & Safe Return to Port (SRtP)

For passenger ships to which SOLAS SRtP applies (broadly, ships ≥ 120 m in length or having three or more main vertical fire zones), the ship must retain, after a defined casualty (fire in one zone or flooding of one watertight compartment), a set of **essential systems** — **including**

propulsion and steering. Where thrusters/pods/steering thrusters form part of the propulsion or steering means, the surveyor must confirm:

- redundancy and physical/functional separation so a single casualty does not disable all propulsion & steering;
- power supply, control and cooling for the retained thruster(s) surviving the casualty (fire zone / flooded compartment);
- consistency with the ship's SRtP systems-availability documentation and casualty-threshold analysis.

This typically *adds* redundancy/segregation drawings and a systems availability analysis to the base thruster package. (*Ref. SOLAS II-2 Reg.21/22; NR467 passenger-ship requirements.*)

15.4 Tankers & Gas Carriers — Hazardous Areas & Explosion Protection

On oil tankers, chemical tankers and gas carriers, the bow-thruster room and any electrical/rotating equipment located in or adjacent to **hazardous (gas-dangerous) zones** must meet explosion-protection requirements. The surveyor should verify:

- hazardous-area classification of the thruster space and boundaries;
- **certified safe-type (Ex) equipment** for motors, sensors and junction boxes where required, with certificates;
- cable entries, penetrations and ventilation preserving zone integrity;
- for gas carriers, compliance with the **IGC Code** zoning in addition to SOLAS II-2.

(*Ref. SOLAS II-2; IEC 60079 series; IGC Code; NR467 Pt C electrical in hazardous areas.*) A tanker bow thruster is often electric; confirm the motor's location relative to the cargo/hazardous zone *before* accepting standard (non-Ex) equipment.

15.5 Offshore & DP-Intensive Vessels

OSVs, construction, cable/pipe-lay units, diving support vessels, lifting units and MODUs are defined by their DP capability. Beyond the DP package of Chapter 12, expect: multiple independent thruster trains, power-plant segregation, DP-3 fire/flood separation where notated, worst-case-failure design intent, and comprehensive FMEA and proving-trials programmes. Diving support vessels (NR467 Pt E, Ch.7) additionally require the thruster/DP package to be reviewed alongside the diving system's own classification (**NR610**) where the two interact (e.g. shared power, station-keeping during diving operations). Mobile offshore units are additionally governed by **offshore-unit / MODU rules**, which may modify hull, machinery and location (hazardous-area) requirements relative to NR467.

15.6 Tugs & Escort Tugs

For tugs the thruster *is* the main propulsion and steering means (Z-drive or VSP). Escort tugs generate very high **indirect steering and towing loads**; the surveyor must confirm the azimuth/steering system, unit structure and hull foundation are substantiated for these loads (not merely free-running thrust), together with the full steering-gear package (Chapter 9) and stability interaction. (*Ref. NR467 Pt D tug / escort-tug service notations.*)

15.7 Thruster Compartment (All Ship Types)

Independent of ship type, the thruster room / bow-thruster compartment carries its own requirements the surveyor should not overlook: watertight boundaries and subdivision, bilge/drainage, fire protection and boundary insulation, ventilation and (for tankers/gas carriers) hazardous-area integrity, plus safe access and escape. Confirm these are reflected in the general arrangement and hull/piping documentation. (*Ref. NR467 Part B, Ch 5; Part C fire & piping; SOLAS II-2.*)

Surveyor note (ship type): Read the *requested notation string* first — it tells you the ship type, service and additional notations, and therefore which overlays of Chapters 8–14 and which *separate rule sets* (SRtP, hazardous-area/IGC, offshore/MODU, naval, HSC) apply. The most common ship-type gaps are: SRtP redundancy on passenger ships, Ex/hazardous-area equipment on tankers and gas carriers, and DP-3 separation on offshore units.

Chapter 16

Testing & Commissioning Requirements

16.1 Factory Acceptance Tests (FAT)

Required for All Thrusters:

- Dimensional verification
- Material verification
- Non-destructive testing
- Assembly inspection
- Mechanical running test
- Hydraulic system test (if applicable)
- Electric motor test (if applicable)
- Control system test
- Retraction mechanism test (retractable)
- CP mechanism test (CP propellers)
- Azimuth slewing test (azimuth thrusters)

16.2 Installation Tests

On-Site Verification:

- Foundation inspection
- Alignment verification
- Piping inspection
- Electrical installation inspection
- Control system integration test
- Hydraulic system commissioning

16.3 Dock Trials

Pre-Sea Trials:

- Static thrust test
- No-load running test
- Control system functional test
- Retraction/extension test (retractable)
- Azimuth rotation test (azimuth)
- CP pitch control test (CP propellers)

16.4 Sea Trials

Performance Verification:

- Performance verification
- Maneuverability trials
- Vibration measurements
- Noise measurements
- Thermal performance
- Steering gear trials (if steering function)
- DP trials (for DP systems)
- DP FMEA proving trials (for DP systems)

16.5 Documentation & Reporting

Required Reports (FI):

- FAT Report
- Installation Report
- Sea Trial Report
- DP Trials Report (for DP systems)
- As-Built Documentation

Chapter 17

Surveyor Checklists by Thruster Type

17.1 Tunnel Thrusters (Fixed) - Complete Checklist

17.1.1 General Information

- ☐ Thruster power rating ≥ 110 kW (Section 15 applies)
- ☐ Thruster manufacturer and model
- ☐ Installation location (bow/stern)

17.1.2 Structural

- ☐ General Arrangement Drawing (**FA**)
- ☐ Tunnel Structural Arrangement (**FA**)
- ☐ Tunnel Structural Calculations (**FA**)
- ☐ Tunnel End Grating/Grid Design (**FA**)
- ☐ Hull Penetration Details (**FA**)

17.1.3 Thruster Unit - Mechanical

- ☐ Thruster Unit General Assembly (**FA**)
- ☐ Propeller Design Drawing (**FA**)
- ☐ Propeller Strength Calculations (**FA**)
- ☐ Propeller Shaft Drawing (**FA**)
- ☐ Shaft Strength Calculations (**FA**)
- ☐ Bearing Arrangement Drawing (**FA**)
- ☐ Bearing Load Calculations (**FA**)
- ☐ Gearbox Assembly Drawing (**FA**)
- ☐ Gear Tooth Calculations (**FA**)
- ☐ Gearbox Lubrication System (**FA**)

17.1.4 Drive System

If Electric:

- ☐ Electric Motor Specification (**FA**)
- ☐ Motor Foundation Drawing (**FA**)
- ☐ Motor-Gearbox Coupling (**FA**)

If Hydraulic:

- ☐ Hydraulic Motor Specification (**FA**)
- ☐ Hydraulic System Schematic (**FA**)
- ☐ See Chapter 11 for complete hydraulic requirements

17.1.5 Control System

- ☐ Control System Block Diagram (**FA**)
- ☐ Control Panel Layout (**FA**)
- ☐ Electrical Single-Line Diagram (**FA**)
- ☐ Control Software Description (**FA**)

17.1.6 Calculations

- ☐ Thrust Calculation (**FA**)
- ☐ Vibration Analysis (**FA**)
- ☐ Noise Level Prediction (**FA**)
- ☐ Thermal Analysis (**FA**)

17.1.7 Materials & Welding

- ☐ Material Specifications (**FA**)
- ☐ Welding Procedures (**FA**)
- ☐ Material Certificates (during construction)
- ☐ NDT Plan (**FA**)

17.1.8 Testing

- ☐ FAT Procedures (**FA**)
- ☐ Installation Test Procedures (**FA**)
- ☐ Sea Trial Procedures (**FA**)
- ☐ Test Reports (**FI** - after testing)

17.1.9 Manuals

- ☐ Operation & Maintenance Manual (FI)
- ☐ Spare Parts List (FI)
- ☐ Installation Manual (FI)

17.2 Retractable Thrusters - Complete Checklist

All items from Section 10.1 (Tunnel Thrusters), PLUS:

17.2.1 Retraction Mechanism

- ☐ Retraction Mechanism Assembly (FA)
- ☐ Hydraulic Cylinder Specifications (FA)
- ☐ Retraction System Calculations (FA)
- ☐ Hull Opening & Closure Design (FA)
- ☐ Hull Opening Structural Analysis (FA)
- ☐ Watertight Integrity Documentation (FA)
- ☐ Locking System Design (FA)
- ☐ Position Indication System (FA)
- ☐ Emergency Lowering System (FA)
- ☐ Hydraulic System for Retraction (FA)

17.2.2 Additional Manuals

- ☐ Retraction/Extension Procedures (FI)
- ☐ Emergency Recovery Procedures (FI)

17.2.3 Additional Testing

- ☐ Retraction mechanism FAT
- ☐ Seal integrity testing
- ☐ Emergency lowering test
- ☐ Dock trials - retraction/extension
- ☐ Sea trials - retraction under way (if applicable)

17.3 Azimuth Thrusters (Fixed Pitch) - Complete Checklist

17.3.1 General & Structure

- ☐ General Arrangement Drawing incl. 360° clearance envelope (**FA**)
- ☐ Azimuth Unit Complete Assembly (**FA**)
- ☐ Hull Penetration & Foundation (**FA**)
- ☐ Hull Structure Calculations / FEA (**FA**)

17.3.2 Propeller & Shaft

- ☐ Propeller Design Drawing (**FA**)
- ☐ Propeller Strength Calculations (**FA**)
- ☐ Vertical Shaft Assembly & Calculations (**FA**)
- ☐ Thrust Bearing Design & Calculations (**FA**)
- ☐ Radial Bearing Arrangement (**FA**)
- ☐ Column Structure Design (**FA**)
- ☐ Seal System Design (**FA**)

17.3.3 Gear & Slewing System

- ☐ Gear Train Assembly (**FA**)
- ☐ Gear Tooth Strength Calculations — ISO 6336 (**FA**)
- ☐ Gear Lubrication System (**FA**)
- ☐ Slewing Bearing Design & Calculations (**FA**)
- ☐ Azimuth Drive Mechanism & Calculations (**FA**)
- ☐ Position Feedback System (**FA**)

17.3.4 Drive System & Prime Mover

- ☐ Electric or hydraulic power-transmission package (**FA**)
- ☐ Slip ring / rotary joint (electric) (**FA**)
- ☐ **Prime-mover add-on** — electric or diesel (Chapter 8)

17.3.5 Control, Calculations, Materials & Testing

- ☐ Control System Architecture & Bridge Console (**FA**)
- ☐ Local Control Station (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Material Specifications, Welding, NDT Plan (**FA**)
- ☐ FAT / dock trial (azimuth slewing) / sea-trial procedures (**FA**)
- ☐ O&M and Installation Manuals (**FI**)

17.3.6 If Steering / DP / Ice / Notations

- ☐ If steering function → Chapter 9 package (SOLAS II-1/29, FMEA)
- ☐ If part of main shaft line → Chapter 10 package
- ☐ If DP → Section 17. (DP add-on below)
- ☐ If ice class → Chapter 13 add-on
- ☐ Overlay applicable notations → Chapter 14

17.4 Azimuth Thrusters (Controllable Pitch) - Complete Checklist

All items from Azimuth FP, PLUS:

- ☐ CP Hub Mechanism Assembly (**FA**)
- ☐ Blade Retention Calculations (**FA**)
- ☐ Pitch Control Mechanism (**FA**)
- ☐ Pitch Control Hydraulic System incl. rotary joint (**FA**) — Chapter 11
- ☐ Pitch feedback & fail-safe position (**FA**)

17.5 Podded Propulsion Units - Complete Checklist

All items from Azimuth (FP or CP), PLUS:

- ☐ Pod-specific structural components & strut (**FA**)
- ☐ Electric motor in pod — specification, cooling, monitoring (**FA**)
- ☐ Power supply through strut / slip-ring (**FA**)
- ☐ Propeller direct-drive arrangement (**FA**)
- ☐ Pod-specific FAT/testing (**FA**)
- ☐ **Electric prime-mover add-on** (Chapter 8)

17.6 Voith-Schneider (Cycloidal) Units - Complete Checklist

- ☐ VSP Unit General Assembly (**FA**)
- ☐ Blade (wing) design + blade-shaft/blade strength calculations (**FA**)
- ☐ Blade bearing arrangement (**FA**)
- ☐ Kinematic control gear / steering mechanism strength (**FA**)
- ☐ Rotor casing & main bevel gear + gear calculations (**FA**)
- ☐ Main thrust bearing design & life (**FA**)
- ☐ Bottom cover / guard plate structure (**FA**)
- ☐ Hull seat, foundation & well structure / FEA (**FA**)
- ☐ Seal system (rotor & blade shafts) (**FA**)
- ☐ Servo / hydraulic control unit (**FA**) — Chapter 11
- ☐ Steering control system & bridge console (**FA**)
- ☐ **Steering-function package** — Chapter 9 (FMEA, redundancy, SOLAS II-1/29)
- ☐ Prime-mover add-on (Chapter 8) + drive-train TVC

17.7 Water-Jet Systems - Complete Checklist

- ☐ Water-jet general arrangement & installation (**FA**)
- ☐ Impeller/rotor design & strength calculations (**FA**)
- ☐ Shaft line & bearings (**FA**) — Chapter 10
- ☐ Inlet duct & transom structure / hull penetration (**FA**)
- ☐ Steering & reversing (bucket) mechanism (**FA**)
- ☐ Steering/reversing hydraulic system (**FA**) — Chapter 11
- ☐ Control system & bridge interface (**FA**)
- ☐ Drive package (direct-from-engine or via gearbox) + TVC (**FA**)
- ☐ Performance / thrust calculations (**FA**)
- ☐ Materials, welding, NDT (**FA**)
- ☐ FAT / trials procedures (**FA**); O&M manuals (**FI**)
- ☐ If steering means of ship → Chapter 9; if DP → DP add-on

17.8 Prime-Mover Add-On Checklist (apply to any type)

If **ELECTRIC**-motor driven:

- ☐ Drive-motor data sheet (**FA**)
- ☐ Frequency converter (VFD) specification & arrangement (**FA**)
- ☐ Harmonic / power-quality analysis (**FA**)
- ☐ Supply transformer & switchboard interface (**FA**)
- ☐ Power cable sizing, routing & segregation (**FA**)
- ☐ Motor protection, monitoring & cooling (**FA**)

If **DIESEL**-engine driven:

- ☐ Diesel engine approval / type data (**FA**)
- ☐ **Torsional vibration calculation of the complete train** (**FA**)
- ☐ Clutch & flexible coupling (**FA**)
- ☐ Engine foundation & chocking (**FA**)
- ☐ Fuel oil / lube oil / cooling water systems (**FA**)
- ☐ Starting system & exhaust gas system (**FA**)
- ☐ Engine control, safety & monitoring (**FA**)
- ☐ NOx Technical File / EIAPP (**FI**)

17.9 Ice Class / Winterisation Add-On Checklist (apply if notation carried)

- ☐ Ice-class notation basis & design conditions (**FA**)
- ☐ Propeller ice-load strength calculation (**FA**)
- ☐ Shaft line & ice-torque verification (**FA**)
- ☐ Gear & azimuth/nozzle ice-load substantiation (**FA**)
- ☐ Thruster lower-unit ice protection (**FA**)
- ☐ Tunnel/inlet grid ice arrangement (**FA**)
- ☐ Winterisation of hydraulic/cooling/seal systems (**FA**)
- ☐ Low-temperature material certification (**FA**)

17.10 DP Thrusters - Additional Checklist Items

For ANY thruster type used in DP system, add:

- ☐ DP System General Arrangement (**FA**)
- ☐ Thruster Configuration Drawing (**FA**)
- ☐ DP Capability Plot (**FA**)
- ☐ FMEA Documentation (**FA**)
- ☐ Redundancy Concept (**FA**)
- ☐ Power Supply Segregation (**FA**)
- ☐ DP Control System Architecture (**FA**)
- ☐ Position Reference System Configuration (**FA**)
- ☐ PMS Configuration (**FA**)
- ☐ UPS Configuration (**FA**)
- ☐ DP FMEA Proving Trials Plan (**FA**)
- ☐ DP Trials Reports (**FI**)
- ☐ DP Operations Manual (**FI**)

Chapter 18

Submission Cover Sheets by Ship Type

How to use: one cover sheet per page. Pick the sheet matching the vessel’s service notation, complete the identification block, then tick each line as the corresponding drawing/document is received and found acceptable. Every sheet shares a *common core* (base package, type-specific package, prime mover, testing, manuals) and then lists the **ship-type-specific extras** — the items most often missing for that service. Use together with the detailed checklists of Chapter 17 and the overlays of Chapters 8–15. Legend: **FA** = For Approval, **FI** = For Information.

18.1 Generic Template (any ship / any thruster type)

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

Core package (all thrusters ≥ 110 kW):

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Functional overlays — tick those applicable:

- ☐ Steering function → Chapter 9 (FMEA, redundancy, SOLAS II-1/29)
- ☐ Part of main shaft line → Chapter 10 (+ MON-SHAFT, Section 10.3, if notated)
- ☐ Dynamic positioning → Chapter 12 (FMEA, capability plot, proving trials)
- ☐ Hydraulic actuation → Chapter 11

Additional-notation & ice overlays — tick those in the notation string:

- ☐ Ice class / Polar / COLD → Chapter 13
- ☐ DP / COMF / AUT / CLEANSHIP / RP / MON-SHAFT → Chapter 14

18.2 Harbour / Escort Tug

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (tug):

- ☐ Azimuth (Z-drive) or Voith-Schneider unit — full type package (**FA**)
- ☐ **Full steering-gear package** — Chapter 9: FMEA, redundancy, control stations, response time, SOLAS II-1/29 (**FA**)
- ☐ **Escort / indirect-towing loads** on steering & structure; unit & foundation substantiation for these loads, not free-running only (**FA**)
- ☐ Uprated slewing duty & brake/locking; stability interaction documentation (**FA**)
- ☐ Drive-train torsional vibration calculation (diesel Z-drive) (**FA**)

18.3 Offshore Supply / PSV / AHTS (DP)

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (OSV / DP):

- ☐ Bow tunnel(s) + retractable + stern azimuth — type packages (**FA**)
- ☐ **DYNAPOS / DP (usually DP-2)** — Chapter 12: FMEA, redundancy concept, power segregation, capability plot, thruster allocation, proving-trials plan (**FA**)
- ☐ Retractable/take-me-home thruster package incl. emergency lowering (**FA**)
- ☐ FiFi and other service systems interfaces where fitted (**FI**)
- ☐ Steering-function compliance for azimuth units (**FA**)

18.4 Passenger Ship / RoPax / Ferry

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (passenger):

- ☐ Bow (+ stern) tunnel thrusters / azimuth — type packages (**FA**)
- ☐ **Safe Return to Port** redundancy & segregation of propulsion/steering (ships ≥ 120 m or 3 fire zones); systems-availability analysis (**FA**)
- ☐ SOLAS steering-gear compliance (II-1/29) where thrusters steer (**FA**)
- ☐ COMF-NOISE / COMF-VIB noise & vibration prediction (**FA**)
- ☐ Emergency/standby power to retained thruster after casualty (**FA**)

18.5 Cruise Ship

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (cruise):

- ☐ Podded propulsors + multiple bow/stern tunnels — type packages (**FA**)
- ☐ **Safe Return to Port** redundancy & systems availability (**FA**)
- ☐ **COMF-NOISE / COMF-VIB** — blade-rate excitation, cavitation noise, resilient mounting (**FA**)
- ☐ Pod electrical-machine, cooling & monitoring; slip-ring/power-through-strut (**FA**)
- ☐ DP package if DP-notated (**FA**)

18.6 Double-Ended Ferry

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (double-ended):

- ☐ Azimuth or Voith-Schneider unit *at each end* — type packages (**FA**)
- ☐ **Steering redundancy at both ends** + control-station changeover logic (**FA**)
- ☐ Response time / manoeuvring in both running directions (**FA**)
- ☐ COMF noise & vibration (passenger comfort) (**FA**)

18.7 Dredger (TSHD / CSD)

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (dredger):

- ☐ Bow tunnels / azimuth — type packages (**FA**)
- ☐ DREDGING service duty; robust manoeuvring provisions (**FA**)
- ☐ DP package where station-keeping/tracking notated (**FA**)
- ☐ Duty-cycle / abrasion considerations in bearing & seal life (**FI**)

18.8 Cable / Pipe Layer & Offshore Construction (DP)

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (construction / DP):

- ☐ Retractable + azimuth + tunnel thrusters — type packages (**FA**)
- ☐ **DYNAPOS DP-2 / DP-3** — Chapter 12: FMEA, redundancy, power segregation, capability plot, proving trials (**FA**)
- ☐ DP-3 **fire / flood zone separation** of thruster groups & power (**FA**)
- ☐ Worst-case-failure design intent documentation (**FA**)

18.9 Drillship / Semi-Submersible (MODU)

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (MODU / DP-3):

- ☐ Multiple azimuth thrusters — type packages (**FA**)
- ☐ **DYNAPOS DP-3** station keeping — full FMEA, redundancy, proving trials (**FA**)
- ☐ **Fire & flood separation** of thruster/power groups (DP-3) (**FA**)
- ☐ Compliance with **offshore-unit / MODU rules** (hull, machinery, hazardous-area) (**FA**)

18.10 Oil / Chemical Tanker

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (tanker):

- ☐ Bow thruster (usually electric) — type package (**FA**)
- ☐ **Hazardous-area classification** of thruster room & boundaries (**FA**)
- ☐ **Certified safe-type (Ex) equipment** for motor/sensors/junction boxes where in or adjacent to dangerous zones, with certificates (**FA**)
- ☐ Cable entries, penetrations & ventilation preserving zone integrity (**FA**)
- ☐ Confirm motor location vs. hazardous zone *before* accepting non-Ex equipment (**FA**)

18.11 Gas Carrier (LNG / LPG)

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (gas carrier):

- ☐ Bow thruster / azimuth — type package (**FA**)
- ☐ **IGC Code** hazardous-zone requirements in addition to SOLAS II-2 (**FA**)
- ☐ Gas-safe / certified (Ex) electrical equipment & certificates (**FA**)
- ☐ Hazardous-area drawings for thruster space & boundaries (**FA**)

18.12 Ice-Class / Polar Trader

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (ice) — Chapter 13:

- ☐ Ice-class notation basis & design conditions (**FA**)
- ☐ **Propeller ice-load strength** calculation (**FA**)
- ☐ **Shaft line & gear ice-torque** verification (blade-milling / blade-loss) (**FA**)
- ☐ Azimuth/pod lower-unit & slewing gear vs. ice-block impact (**FA**)
- ☐ Winterisation (low-temp oils, materials, sensor heating; anti-icing) (**FA**)
- ☐ Low-temperature material certification (**FA**)

18.13 Research / Naval / Patrol

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (research / naval):

- ☐ Rim-driven / podded / azimuth units — type packages (**FA**)
- ☐ **Low underwater-noise / signature** & COMF (silent) documentation (**FA**)
- ☐ Cavitation-noise control measures (**FA**)
- ☐ Naval vessels: compliance with applicable **naval ship rules** (**FA**)

18.14 Large Yacht

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (yacht):

- ☐ Bow/stern tunnel thrusters and/or pods — type packages (**FA**)
- ☐ **Comfort** (noise & vibration) documentation (**FA**)
- ☐ Retractable-thruster & stabiliser interface where fitted (**FA**)
- ☐ Applicable yacht-specific rules (**FA**)

18.15 Container / Bulk / General Cargo

Vessel / Yard No		Class notation	
IMO number		Thruster type / qty	
Prime mover		Reviewer / date	

- ☐ General Arrangement + Technical Specification sheet (**FA**)
- ☐ **Type-specific mechanical/structural package** — attach the matching Chapter 17 checklist (**FA**)
- ☐ Materials, welding procedures, NDT plan (**FA**)
- ☐ Thrust / vibration / noise / thermal calculations (**FA**)
- ☐ Control system package + electrical single-line diagram (**FA**)
- ☐ **Prime-mover package** (Chapter 8): if *electric* — motor + VFD + harmonic study + cabling/segregation + protection; if *diesel* — engine approval + **full-train TVC** + clutch/coupling + FO/LO/cooling/starting/exhaust + controls (**FA**)
- ☐ FAT / dock-trial / sea-trial procedures (**FA**); test reports (**FI**)
- ☐ O&M, spare-parts & installation manuals (**FI**)

Ship-type-specific extras (cargo):

- ☐ Bow tunnel thruster for manoeuvring — type package (**FA**)
- ☐ Typically electric drive — confirm motor, cabling & protection (**FA**)
- ☐ No special service overlay unless ice class / other notation carried (verify string) (**FA**)

Chapter 19

Common Submission Errors & Surveyor Notes

19.1 Frequent Documentation Gaps

19.1.1 Missing or Incomplete Calculations

- Shaft strength calculations missing fatigue analysis
- Bearing life calculations without proper load factors
- Gear calculations not per recognized standards (ISO 6336)
- Vibration analysis incomplete or missing
- Hydraulic system calculations (accumulator sizing, pressure drop)

19.1.2 Inadequate Drawings

- General arrangements without sufficient detail
- Material specifications not clearly indicated
- Welding details missing or incomplete
- Foundation drawings without load specifications
- Control system diagrams without protection details

19.1.3 Steering Gear Overlap Not Addressed

- Azimuth or podded thrusters used for steering without steering gear compliance documentation
- Missing FMEA for steering function
- Inadequate redundancy analysis
- Missing emergency steering provisions
- SOLAS compliance not demonstrated

19.1.4 DP System Gaps

- FMEA not covering all failure modes or worst-case scenarios
- Inadequate power segregation for DP-2/DP-3
- Missing fire/flood zone analysis for DP-3
- Thruster interaction effects not analyzed
- Incomplete proving trials procedures

19.1.5 Hydraulic System Issues

- Undersized accumulators
- Inadequate pressure relief provisions
- Missing filtration specifications
- Incomplete piping stress analysis
- Emergency provisions not clearly defined

19.1.6 Prime-Mover Gaps

- Diesel-driven Z-drive submitted *without* a torsional vibration calculation for the complete engine–clutch–shaft–thruster train (most common tug/OSV gap)
- Missing barred-speed range or clutch-in/misfiring conditions in the TVC
- Electric-driven thruster without harmonic / power-quality study or cable-sizing & segregation
- Hybrid PTI/PTO arrangements with undocumented driving paths or mode-changeover logic

19.1.7 Ice-Class & Notation Gaps

- Ice-classed vessel with propeller/shaft/gear checked only for open-water rating, not ice peak-torque / blade-milling loads
- Azimuth/pod lower unit and slewing gear not substantiated against ice-block impact
- Winterisation (low-temperature oils, materials, sensor heating) not addressed
- Comfort (COMF) notation carried but no thruster noise/vibration prediction
- Requested notation string not reconciled with the submitted thruster package

19.1.8 Cycloidal (VSP) & Combined Propulsion/Steering Gaps

- VSP mechanical package submitted without the steering-function compliance set (FMEA, redundancy, SOLAS II-1/29, response time)
- Blade guard/bottom plate and its hull attachment not substantiated for grounding/impact
- Steerable water-jet / nozzle / grid units missing steering-gear documentation

19.2 Surveyor Review Tips

19.2.1 Initial Review

1. Verify thruster power ≥ 110 kW → Section 15 applies
2. Identify thruster type → Apply appropriate checklist (Chapters 3–7)
3. Identify the prime mover (electric / diesel / hydraulic) → Add Chapter 8 submissions
4. Check for steering function → Add Chapter 9 requirements
5. Check for main shafting classification → Add Chapter 10 requirements
6. Check for DP application → Add Chapter 12 requirements
7. Check ice class / winterisation notation → Add Chapter 13 requirements
8. Reconcile the full class-notation string → Add Chapter 14 requirements

19.2.2 Calculation Review

- Verify all calculations use BV-approved formulas or recognized standards
- Check safety factors applied
- Verify material properties used in calculations match specifications
- Ensure all loading conditions considered (normal, extreme, emergency)

19.2.3 Drawing Review

- Check for consistency between drawings
- Verify dimensions and tolerances
- Ensure material specifications clearly indicated
- Check for proper welding symbols and details
- Verify accessibility for maintenance

19.3 Red Flags for Surveyors

19.3.1 Immediate Rejection Criteria

- Missing critical calculations (shaft strength, bearing life, gear strength)
- Material specifications not compliant with BV rules
- Welding procedures not approved
- Control system without adequate safety provisions
- Steering gear requirements not addressed when applicable
- DP FMEA incomplete or inadequate

19.3.2 Require Clarification

- Ambiguous material specifications
- Incomplete or unclear drawings
- Calculations without clear assumptions
- Testing procedures without acceptance criteria
- Manuals not in working language of crew

Chapter 20

Quick Reference Tables

20.1 Thruster Type vs. Applicable Rule Sections

Table 20.1: Applicability Matrix

Thruster Type	Sec.15	Sec.14	Sec.7	DP (Pt F)	Hydraulics
Tunnel (Fixed)	✓	Optional	Optional	Optional	If hydraulic
Retractable	✓	Optional	Optional	Optional	✓ Required
Azimuth FP	✓	Often ✓	Typically ✓	Optional	If hydraulic
Azimuth CP	✓	Often ✓	Typically ✓	Optional	✓ Required
Podded	✓	Often ✓	Typically ✓	Optional	If CP
Water-Jet	✓	Often ✓	✓ Required	Optional	✓ Required
Voith-Schneider	✓	✓ Required	Typically ✓	Optional	✓ Required
Rim-Driven	✓	Optional	Optional	Optional	Rarely
Pump-Jet / Ducted	✓	If steering	✓ Required	Optional	If steering
Gill / Hydro-Jet	✓	Optional	Optional	Optional	If hydraulic
Steering Grid / Nozzle	✓	✓ Required	✓ Required	Optional	✓ Required

Sec.15 = Thrusters; Sec.14 = Steering Gear; Sec.7 = Shaft lines/propellers (NR467 Pt C, Ch 1). “Often/Typically ✓” means applicable when the unit provides steering or forms part of the main shaft line. Overlay Chapter 8 (prime mover), Chapter 13 (ice class) and Chapter 14 (notations) on every row as applicable.

20.2 Prime Mover & Additional-Notation Quick Guide

Table 20.2: Prime Mover — Key Additional Submissions

Prime mover	Headline additional submissions (see Chapter)
Electric motor	Motor data sheet; frequency converter (VFD); harmonic/power-quality study; transformer & switchboard interface; cable sizing & segregation; motor protection/monitoring/cooling.
Diesel engine	Engine approval (Sec.2); full-train torsional vibration calculation ; clutch & flexible coupling; foundation/chocking; FO/LO/cooling/starting/exhaust systems; engine safety & controls; NOx file.
Hydraulic motor	Per Chapter 11 (HPU, circuit, accumulators, relief, filtration, monitoring).

Table 20.3: Additional Notations — Trigger Reference

Notation carried	Adds
DP / DYNAPOS	Chapter 12: FMEA, redundancy, power segregation, capability plot, proving trials (NR467 Pt F, DYNAPOS).
Ice class / Polar / COLD	Chapter 13: ice-load propeller/shaft/gear strengthening, winterisation, low-temp materials (NR467 Pt E, NR527, IACS UR I3).
COMF-NOISE / COMF-VIB	Chapter 14: noise & vibration prediction, blade-rate excitation, resilient mounting.
AUT-UMS / AUT-CCS	Chapter 14: extended alarms/monitoring, safe remote control, fail-safe.
CLEANSHIP	Chapter 14: EAL seal/hydraulic fluids, leakage monitoring.
MON-SHAFT	Section 10.3: oil-lubricated bearing/seal, lube-oil analysis & wear-down monitoring, condition-based shaft survey.
TUG / ESCORT TUG	Chapter 14: uprated steering & structural loads.

20.3 Document Approval Categories Summary

Table 20.4: Approval Categories

Category	Meaning	Timing	Surveyor Action
FA	Must be reviewed and approved before construction	Submit before construction	Detailed technical review
FI	Submitted for record, not requiring formal approval	Submit as specified	Review for completeness
During Construction	Certificates/reports generated during manufacturing	Submit as work progresses	Verify during surveys

Chapter 21

Regulatory References

21.1 Bureau Veritas Rules

21.1.1 NR467 - Rules for Classification of Steel Ships

Part A: Classification and Surveys — Chapter 1 (“Principles of Classification and Class Notations”) is where every **service notation** and **additional class notation** is formally *listed and defined* (the master index of ship types and notations), together with the principles of assignment/maintenance of class (Ch.2) and survey scope (Ch.3–5). The *detailed technical requirements* for each notation then sit in Parts D/E (service notations) and F (additional class notations) as set out below — Part A tells the surveyor *which* notations exist and apply; Parts D/E/F tell the surveyor *what to demand* for each one.

Part B: Hull and Stability

- Chapter 2: Subdivision and Stability
- Chapter 3: Materials
- Chapter 4: Welding
- Chapter 5: Hull Structure

Part C: Machinery, Electricity, Automation and Fire Protection

- Chapter 1 — Machinery:
 - Section 2: Diesel Engines (prime-mover-driven thrusters)
 - Sections 7–8: Shaft Lines and Propellers
 - Section 14: Steering Gear
 - Section 15: Thrusters (and water-jets)
- Chapter 2 — Electrical Installations (drive motors, converters, cables)
- Chapter 3 — Automation (control, alarms, AUT notations)
- Chapter 4 — Fire Protection

Note: exact section numbering varies with rule edition — always confirm against the current NR467 in Rules Explorer.

Part D: Service Notations — general ship-type service notations (e.g. passenger ship, oil tanker, dredger, container ship). Determines the ship-type requirements overlaid on the thruster package.

Part E: Service Notations for Offshore Service Vessels and Tugs — e.g. *tug*, *escort tug*, *supply*, anchor-handling. Source of the uprated escort/towing steering & structural loads for tug/OSV thrusters.

Part F: Additional Class Notations — the additional (voluntary) notations, in multiple chapters:

- Ice class (IA Super / IA / IB / IC) and Polar Class (PC1–PC7); COLD winterisation
- Comfort on board — COMF-NOISE, COMF-VIB
- Automated machinery spaces / integrated systems — AUT-UMS, AUT-CCS, AUT-PORT
- Dynamic positioning — **DYNAPOS** (AM / AT / AM-AT / AM-AT R / AM-AT RS / SAM)
- Environmental protection — CLEANSHIP and related (EAL lubricants)
- Redundant propulsion — **AVM-DPS** (Part F, Ch.2); related **HEADING CONTROL-DS** / **HEADING CONTROL-IS** (Part F, Ch.11)
- Shaft-line monitoring — MON-SHAFT (Part F, Ch.5)
- Electric propulsion / power supply (Part F, Ch.14) — relevant to fully electric or hybrid pod/azimuth thruster installations; cross-check against Chapter 8
- Elastic shaft alignment & hydroelasticity (Part F, Ch.15) — relevant where a thruster’s vertical/propeller shaft alignment scheme claims an elastic-alignment notation; cross-check against Chapter 10

Reference correction: Dynamic positioning is an *additional class notation* (**DYNAPOS**) defined in **NR467 Part F**. There is no “NR216 DP rule”: **NR216** is the *Rules on Materials and Welding* (see below). Earlier drafts of this catalog mis-cited NR216 for DP — read DP references as **NR467 Part F (DYNAPOS)**.

21.1.2 Related BV Rule Notes (as applicable)

Thruster-specific:

- **NR584 — Azimuth thrusters in ice for polar class ships and icebreakers** (Apr 2026). *The* dedicated azimuth-thruster rule note; controlling document for POLAR CLASS / icebreaker azimuth propulsion. See Section 1.14.

Materials, survey and certification:

- **NR216** — Rules on Materials and Welding for the classification of marine units
- **NR266** — Requirements for Survey of Materials and Equipment
- **NR320** — Certification scheme of materials and equipment for marine units

- **NR480** — Approval of the manufacturing process of metallic materials
Ship / unit type (the governing rule set):
- **NR467** — Rules for the classification of *steel ships* (the default)
- **NR445** — Classification of *offshore units* (MODU); with **NR569** drilling ships and **NR571** column-stabilised units
- **NR483** — Classification of *naval ships*
- **NR500** — Classification and certification of *yachts*
- **NR396** — Classification of *high-speed craft*
- **NR566** — Hull arrangement, stability and systems for *ships* < 500 GT; with **NR600** for hull of small cargo / non-cargo ships
- **NR217** — Classification of *inland navigation vessels*
- **NR645** — Classification of *FSRU and FSU*
- **NR527** — Ships operating in *polar waters* and icebreakers
- **NR610** — Classification of *diving systems*. Per NR467 Pt A, Ch. 1, Sec. 1, [1.1.3], where the requirements differ **NR610 prevails** for the diving system — relevant on a DSV whose DP/thruster package interfaces with it.

Systems and services touching the thruster package:

- **NR659** — Rules on *cyber security* for the classification of marine units
- **NR674** — *Condition monitoring* systems (supports MON-SHAFT / CBM survey)
- **NR632** — *Hardware-in-the-loop* testing (DP and thruster control validation)
- **NR614** — *Underwater radiated noise* (URN)
- **NR598** — Implementation of *safe return to port* and orderly evacuation
- **NR656** — *Power generation units* (power plant feeding electric thrusters)
- **NR526** — *Lifting appliances* onboard ships and offshore units

Guidance notes:

- **NI663** — Propeller in composite materials
- **NI543** — Ice reinforcement selection in different world navigation areas
- **NI565** — Ice characteristics and ice/structure interactions
- **NI649** — Lay-up and reactivation of DP vessels
- **NI640** — Structural assessment of passenger ships and ro-ro ships

21.2 International Standards

21.2.1 ISO Standards

- **ISO 6336**: Calculation of load capacity of spur and helical gears
- **ISO 22547**: Ships and marine technology — azimuth thrusters
- **ISO 484**: Shipbuilding — ship screw propellers (manufacturing tolerances)
- **ISO 4413**: Hydraulic fluid power — general rules for systems
- **ISO 20283 / ISO 6954**: Mechanical vibration — measurement & evaluation (supports COMF-VIB)

21.2.2 IEC Standards

- **IEC 60092 Series**: Electrical installations in ships
 - IEC 60092-101: Definitions and general requirements
 - IEC 60092-201: System design — general
 - IEC 60092-301: Equipment - Generators and motors
 - IEC 60092-352: Choice and installation of cables
 - IEC 60092-504: Control and instrumentation
- **IEC 60034 Series**: Rotating electrical machines (ratings, cooling, temperature rise)
- **IEC 61508**: Functional safety (for DP / thruster control systems)

21.2.3 IACS Unified Requirements & Codes

- **IACS UR I3** — *Machinery Requirements for Polar Class Ships*. Covers loads due to propeller-ice interaction for **azimuthing and fixed thrusters** with geared transmission or an integrated electric motor (geared and podded propulsors). Propulsion machinery type factor $k_3 = 1,0$ for fixed propulsors and **1,2** for azimuthing propulsors. (UR I1/I2 cover the Polar Class descriptions and structural requirements.)
- **IACS UR E26** — *Cyber Resilience of Ships*, and **UR E27** — *Cyber Resilience of On-board Systems and Equipment*. **Mandatory for ships contracted for construction on or after 1 July 2024.**
- **IACS UR E22** — On board use and application of programmable electronic systems
- *Caution: UR M78 is Internal Combustion Engines Fuelled by Gases or Low-flashpoint Fuels* — it is **not** a thruster requirement and should not be cited in thruster work.
- **IMO Polar Code**: Ships operating in polar waters
- **Finnish-Swedish Ice Class Rules (TRAFICOM)**: Baltic ice classes basis

21.2.4 SOLAS Convention

- **Chapter II-1, Regulation 29:** Steering gear
- **Chapter II-1, Reg. 8-1 & II-2, Reg. 21/22:** Safe Return to Port / systems availability after a casualty (passenger ships)
- **Chapter II-2:** Fire protection; hazardous-area boundaries
- **Chapter V, Regulation 28:** Maneuverability

21.2.5 Ship-Type / Service Rule Sets & Codes

- **IEC 60079 series:** Explosion protection — equipment in hazardous areas (tankers, gas carriers, bow-thruster rooms)
- **IGC Code:** Gas carriers — hazardous-zone requirements
- **IMO MODU Code / offshore-unit rules:** Mobile offshore drilling units and offshore service craft
- **HSC Code:** High-speed craft (may modify/replace NR467 machinery requirements)
- **Naval ship rules** (e.g. BV NR483): Naval and patrol vessels — signature/silent operation
- **NR467 Part D:** Service notations (tug, escort tug, supply, dredging, passenger, tanker, etc.)

21.2.6 IMO Resolutions

- **IMO Res.A.468(XII):** Code on Noise Levels
- **IMO Res.MSC.337(91):** Code on Noise Levels on Board Ships (mandatory)
- **IMO MSC/Circ.645 & MSC.1/Circ.1580:** Guidelines for vessels/units with DP systems
- **IMO MSC.1/Circ.1369:** Interim explanatory notes for SRtP assessment

21.3 Online Resources

21.3.1 Bureau Veritas Marine & Offshore

- **Rules Explorer:** <https://ruleexplorer.bureauveritas.com/>
- **NR467 Rules:** <https://marine-offshore.bureauveritas.com/nr467-rules-classification-steel-ships>
- **All Rules & Guidance Notes:** <https://marine-offshore.bureauveritas.com/rule-s-classification-rule-notes-and-guidance-notes>

Chapter 22

Contact Information & Document Control

22.1 For Technical Queries

Bureau Veritas Marine & Offshore Division

- Regional Technical Support Offices
- Email: Contact through official BV channels
- Rules Explorer Support: Available through BV website

22.2 For Urgent Classification Matters

- Contact your assigned Plan Approval Manager
- Regional Marine & Offshore Director

22.3 Document Revision History

Table 22.1: Revision History

Version	Date	Changes	Author
1.0	2025	Initial comprehensive catalog based on NR467 Sections 7, 14, 15 and NR216	Jules (BV AI Assistant)
1.1	2026	Added Voith-Schneider (cycloidal) and other propulsor types (rim-driven, pump-jet, gill/jet, steering grid); prime-mover chapter (diesel-engine vs. electric-motor drive incl. full-train TVC); ice class & winterisation chapter; additional BV notations chapter (COMF, AUT, CLEANSHIP, RP, service notations); expanded checklists, quick-reference tables and standards. Preamble/compile fixes.	Plan Approval Reference (expanded ed.)
1.2	2026	Added Ship-Type & Service-Notation chapter (ship-type → thruster expectation matrix; Safe Return to Port for passenger ships; hazardous-area/Ex for tankers & gas carriers; offshore/MODU & DP-intensive vessels; tugs & escort tugs; thruster-compartment requirements). Added ship-type/service rule sets & codes to references (IEC 60079, IGC, MODU, HSC, naval, SRtP).	Plan Approval Reference (expanded ed.)
1.3	2026	Added MON-SHAFT shaft-line condition-monitoring notation (Section 10.3) with submissions table and cross-references in the notation matrix/quick-guide; added hydraulic-system rule-basis note (Sec.14/Sec.15/piping, ISO 4413, fail-safe pitch). Added per-ship-type submission cover sheets (one-page tick-off) for 15 ship types plus a generic template.	Plan Approval Reference (expanded ed.)
1.4	2026	Readability/accessibility: switched to scalable Latin Modern fonts (crisp, selectable, searchable text) with microtype; enabled navigable PDF bookmarks (open, numbered) and FitH view; added “Navigating this PDF” guide; slightly opened line spacing.	Plan Approval Reference (expanded ed.)
1.5	2026	Rule-reference correction (verified against NR467 Jan. 2026):	Plan Approval Reference (ex

22.4 Important Disclaimers

! CRITICAL NOTICE FOR SURVEYORS:

1. **Verification Required:** This catalog is a reference guide compiled from Bureau Veritas rules. Always verify against the latest official BV rules and guidance notes.
2. **Rule Updates:** BV rules are updated periodically. Ensure you are working with the current version applicable to the vessel's build date and classification notation.
3. **Complex Cases:** For unusual thruster configurations, hybrid systems, or novel technologies, consult with BV technical specialists before approval.
4. **Project-Specific Requirements:** Some projects may have additional requirements beyond standard rules (e.g., owner specifications, flag state requirements, special notations).
5. **Overlapping Requirements:** When thrusters serve multiple functions (propulsion + steering, or DP + steering), ALL applicable requirements must be met.
6. **FMEA — check the notation first:** for DP, the FMEA of Pt F, Ch. 11, Sec. 5, Tab. 1 #17 is required **for the R and RS symbols only** (DYNAPOS AM/AT R, AM/AT RS). Where it applies it is the cornerstone document and an inadequate FMEA is grounds for rejection — but do not demand one for a plain AM / AT / AM-AT installation.
7. **Steering Gear Compliance:** If thrusters provide steering, full SOLAS steering gear requirements apply.
8. **Material Traceability:** All critical components must have traceable material certificates.
9. **Software Validation:** Control system software, especially for DP and steering functions, requires validation documentation.
10. **Cybersecurity:** Modern thruster control systems must address cybersecurity per latest BV guidance and IMO requirements.

! PLEASE CAREFULLY REVIEW THIS DOCUMENT

It may contain errors or omissions. Always verify against official BV rules.

Bureau Veritas

Thruster Requirements Comprehensive Surveyor Catalog

Document Version 1.8
2026

For Bureau Veritas Plan Approval Surveyors

Source Links:

<https://marine-offshore.bureauveritas.com/nr467-rules-classification-steel-ships>
<https://marine-offshore.bureauveritas.com/rules-classification-rule-notes-and-guidance-notes>
<https://rulesexplorer.bureauveritas.com/>